Exploring the Role of Cybersecurity Awareness and Training in Enhancing Sustainable Business Performance in Small and Medium-Sized Enterprises: A Qualitative Case Study

by

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Abstract

Small and medium-sized enterprises (SMEs) are pivotal to global economic growth, innovation, and job creation, yet they face heightened exposure to cybersecurity threats due to limited resources, awareness, and technical capacity. As digital integration advances, these vulnerabilities become more pronounced. This qualitative case study investigates the influence of cybersecurity awareness and training on the sustainable performance of K2 Partnering Solutions, a Boston-based SME in the digital consulting sector. Guided by two central research questions—what cybersecurity strategies are in place to protect organizational data, and how employees are trained to detect and respond to cyber threats—the study draws on data from 15 key stakeholders through semi-structured interviews, online surveys, and document analysis. Thematic analysis and methodological triangulation reveal persistent deficiencies in the customization, implementation, and evaluation of cybersecurity training programs. The findings emphasize the necessity for context-specific, practical awareness initiatives that cultivate a security-conscious workforce capable of mitigating evolving digital risks. Furthermore, the research identifies the absence of robust evaluation frameworks that hinder the assessment and refinement of training effectiveness. By providing a scalable and actionable model tailored to the constraints of SMEs, this study contributes to strengthening digital resilience within resource-limited environments. Its insights not only inform strategic improvements at K2 Partnering Solutions but also offer broader implications for cybersecurity practitioners, policymakers, and SMEs globally striving to safeguard sensitive data, enhance operational resilience, and ensure long-term sustainability in an increasingly hostile cyber landscape.

Keywords: SME cybersecurity, cybersecurity training, digital resilience, cyber risk management, SME cyber defense

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Table of Contents

Abstract **ii**

List of Tables **vii**

SECTION 1: Introduction to Doctoral Research Project **1**

General Problem Statement 2

Specific Problem Statement 3

Purpose Statement 3

Significance of the Study 4

Literature Support 5

Research Questions and Methodology 12

SECTION 2: Literature Support **15**

Description of the Problem and Business Practices 15

Critical Review of Prior Research 16

Justification for a Qualitative Research Design 51

Research Framework 54

Summary of Literature Support 64

SECTION 3: Research Methodology **72**

Role of the Researcher, Background, and Experience 72

Population and Sampling 73

Data Collection and Analysis 82

Assumptions, Limitations, Delimitations 90

SECTION 4: Results, Discussions, and Implications 94

Presentation of Findings 94

Implication of Findings and Recommendations for Future Research 122

Summary and Study Conclusions 135

References 138

Appendix A: Informed Consent Form 174

Appendix B: Interview Questions 178

Appendix C: Survey Questionnaire 181

Appendix D: Document Review Checklist 186

List of Tables

Figure 1. Integrated TAM-TOE Framework.....59

Figure 2. Leadership Involvement 101

Figure 3. Training Frequency.....104

Figure 4. Effectiveness of current training programs 106

Figure 5. Incident Reporting Behavior 107

Figure 6. Perception of Cybersecurity Culture Promotion 112

Figure 7. Cybersecurity's Impact on Long-Term Success 115

Figure 8. Proposed Improvements 117

SECTION 1: Introduction to Doctoral Research Project

Small and medium-sized enterprises (SMEs) are fundamental to the world economy, contributing considerably to development, innovation, and job creation. Representing over 90% of all corporate operations worldwide, SMEs serve as the backbone of economic activity across sectors and regions (Chidukwani et al., 2022). The swift progression of internet technologies, including cloud computing and digital platforms, has accelerated operational efficiencies for SMEs and enhanced their competitiveness in increasingly digital marketplaces. However, this digital transformation has introduced a complex array of cybersecurity challenges disproportionately affecting SMEs. As Yudhiyati et al. (2021) note, integrating digital tools has made SMEs more susceptible to sophisticated cyberattacks, especially given their often-limited technical defenses.

Empirical data underscores the heightened vulnerability of SMEs to cybersecurity threats. A 2018 Ponemon Institute study found that only 28% of SMEs expressed confidence in managing cybersecurity risks, with 67% targeted and 58% experiencing data breaches within the previous year. Alarmingly, nearly half of respondents admitted to lacking a clear understanding of how to protect their systems (Al-Somali et al., 2024). This knowledge gap and growing threat sophistication expose SMEs to significant operational and reputational risks.

The strategic importance of SMEs within supply chains further magnifies the implications of cybersecurity breaches. Many SMEs handle sensitive financial data and personal information on behalf of larger partner organizations, making them attractive entry points for cybercriminals (Arroyabe et al., 2024). A breach at the SME level can compromise the integrity

of SMEs and the larger networks that depend on them, resulting in widespread disruptions and financial losses.

Despite the growing threat landscape, SMEs face persistent hurdles in executing efficient cybersecurity measures. Rawindaran et al. (2023) emphasize that many SMEs underestimate the importance of cybersecurity, often due to limited awareness and restricted financial resources. These limitations make it difficult to justify and sustain cybersecurity infrastructure or talent investments. Moreover, recruiting and retaining skilled cybersecurity professionals remains a significant challenge, leaving many SMEs underprepared to defend against evolving threats.

To ensure long-term viability and protect sensitive data, SMEs must prioritize cybersecurity through organizational awareness and strategic implementation of robust protective measures. As Bada and Nurse (2019) suggest, educating employees about digital risks and enforcing proactive security protocols are essential to building resilience in a digital environment.

General Problem Statement

The general problem to be addressed is the lack of education and awareness about data security practices in small and medium-sized businesses (SMBs), resulting in vulnerabilities to cyberattacks that can compromise sensitive information and disrupt operations.

Supporting Sentences

- Chidukwani et al. (2022) highlighted the growing trend of cyber attackers targeting small and medium-sized businesses (SMBs), highlighting their vulnerability due to their lack of awareness about protecting their systems.

- Junior et al. (2023) noted a significant rise in cyber-attacks targeting SMEs, with 58% of attacks in 2019 directed at them and 43% of breaches involving SMEs as victims. They attributed this increase to the lack of preparedness and awareness of threats among SMEs.

- Raineri and Resig (2020) observed that small businesses are particularly susceptible to cyberattacks due to high cybersecurity expenses, inadequate education, and insufficient training. This unpreparedness makes them appealing targets for cyber attackers.

- Bada and Nurse (2019) observed a significant increase in cybercrime across all business sectors, mainly targeting small and medium-sized enterprises (SMEs). These businesses often have inadequate cybersecurity measures, lacking the awareness, expertise, and resources that larger organizations typically possess.

Specific Problem Statement

The specific problem to be addressed is the lack of education and awareness about data security practices in K2 Partnering Solutions, resulting in vulnerabilities to cyberattacks that can compromise sensitive information and disrupt operations.

Purpose Statement

The purpose of this qualitative study is to explore how cybersecurity awareness and training contribute to the sustainable business performance of K2 Partnering Solutions, a small to medium-sized enterprise (SME) located in Boston, Massachusetts, United States. This research examines the relationship between management and employee awareness of cybersecurity issues

and the effectiveness of training and awareness programs. It aims to uncover how these factors contribute to the company's long-term success and resilience in an increasingly digital landscape.

To attain the research objectives, the researcher will employ qualitative methods, primarily through interviews featuring open-ended questions. This approach facilitates the gathering of detailed responses that align with the study's aims. As noted by Lim (2024), qualitative research is especially well suited for examining multifaceted topics, as it captures the nuances and contextual factors that quantitative methods, which rely on numerical data and statistical analysis, may overlook. By emphasizing the understanding of social phenomena in their natural contexts and meaningfully engaging with participants' experiences, qualitative research proves to be the most suitable method for conducting an in-depth investigation aimed at addressing real-world societal challenges discussed in this study.

Significance of the Study

The significance of this study is grounded in its exploration of the critical relationship between cybersecurity awareness and education and their direct impact on the sustainability of small and medium-sized enterprises (SMEs). Many SMEs remain unaware of online threats and often lack sufficient security measures. As cybersecurity incidents targeting these businesses continue to increase, they risk losing revenue and experiencing a decline in customer trust, ultimately leading to business failure. Understanding how cybersecurity education promotes effective security behavior among employees is crucial for safeguarding against data breaches and fostering a robust IT security culture. The results of this study could offer valuable insights for small and medium-sized enterprises (SMEs) facing growing challenges from cyber threats.

By implementing comprehensive education and awareness programs and strengthening technological defenses, organizational processes, and policy guidelines, businesses can enhance the protection of their digital assets against potential cyber risks.

Literature Support

This literature review analyzes the cybersecurity posture of small and medium-sized enterprises (SMEs). The first section overviews these businesses' challenges and outlines various initiatives to address these issues. It introduces SMEs and their economic significance, explores their adoption of digital technologies, discusses the growing threat of cybercrime, and highlights the factors contributing to their vulnerability to cybercriminals.

This section underscores the vital importance of cybersecurity for small and medium-sized enterprises (SMEs), highlighting awareness and training solutions as effective strategies for risk mitigation. The subsequent section of the literature review thoroughly evaluates scholarly research on cybersecurity awareness and training, concentrating on its evolution, current landscape, emerging trends, and best practices successfully implemented by organizations. The results of this analysis will contribute to designing a cybersecurity strategy specifically suited for K2 Partnering Solutions, allowing the company to protect its digital resources while promoting a robust security awareness culture.

Small and Medium-Sized Enterprises (SMEs)

Small and medium-sized enterprises (SMEs), defined as “firms with 200 or fewer employees” (Bayraktar & Algan, 2019, p. 2), constitute approximately 95% of global enterprises, underscoring their central role in the global economy. Beyond serving as the backbone of

economic activity, SMEs are vital contributors to innovation, employment, and regional development (Bayraktar & Algan, 2019). In the United States, for example, SMEs account for 99.9% of all businesses and employ 60.6 million individuals, representing 47.1% of the private workforce (Downing, 2021). Their economic significance extends into supply chain networks, where they provide critical materials and specialized services to larger firms. In the United Kingdom, 77% of small businesses fulfill such roles, supporting broader industrial productivity and resilience (Federation of Small Businesses [FSB], 2018). SMEs' flexibility and localized expertise enhance supply chain efficiency and drive sustainable economic development and job creation across sectors.

Digital Technology and SMEs

The rapid advancement of digital technologies is a key factor in shaping the modern economic landscape. Innovations such as wireless communication, widespread internet access, mobile applications, and dynamic social media platforms have all contributed to developing a vibrant and inclusive digital economy. As noted by Arroyabe et al. (2024), digital transformation enables small and medium-sized enterprises (SMEs) to achieve a competitive edge by significantly enhancing their operational efficiency and overall performance. These technologies streamline business processes and create new opportunities for market engagement, customer interaction, and data-informed decision-making. However, as SMEs become more dependent on digital tools, many face significant challenges due to limited knowledge and awareness about data security practices. This gap renders businesses susceptible to hacks that may jeopardize critical information, disrupt operations, and erode the gains intended by digitalization.

Cybercrime and Its Growing Threat

Cybercrime has become a major threat to organizational security and sustainability in today's digital economy. Defined as "the use of a computer as an instrument to further illegal ends, such as committing fraud, trafficking in child pornography and intellectual property, stealing identities, or violating privacy" (Dennis, 2024, p. 1), cybercrime presents challenges for enterprises undergoing digital transformation. As small and medium-sized enterprises (SMEs) increasingly adopt emerging technologies to stay competitive, they unintentionally become targets of sophisticated cyber threats (Taskin et al., 2025). Cybercriminals exploit system vulnerabilities through tactics such as malware, phishing, and ransomware, resulting in substantial financial losses, reputational damage, and operational disruptions (Zwilling et al., 2022). Notwithstanding these emergent risks, many SMEs lack the education and awareness to implement adequate data security measures, exposing them to cyberattacks that can compromise sensitive data and disrupt critical business functions.

Globally, the costs associated with cybercrime were anticipated to reach $9.5 trillion in 2024 and escalate to $10.5 trillion by 2025. Specific industries, including healthcare, manufacturing, and finance, remain prime targets. Notably, the healthcare sector has experienced a dramatic 239% increase in breaches over the past four years, with an average cost of $11 million per incident (Arroyabe et al., 2024).

Cybercrime and SMEs

The 2019 Verizon Data Breach Investigations Report reveals that small and medium-sized enterprises (SMEs) are responsible for 43% of all data breaches. SMEs are particularly

vulnerable to cybercriminals due to their access to sensitive customer and employee data, which can be exploited for financial fraud and identity theft (Teymourlouei, 2018). Cybersecurity breaches compromise the confidentiality, integrity, and availability of data, resulting in reputational harm, diminished customer trust, and financial instability (Johnson, 2019). An IBM report states that SMEs incur an average cost of $3.86 million per data breach, with phishing and social engineering attacks being the most common threats (Cost of a Data Breach Report 2020, 2020).

Why SMEs Are Prime Targets for Cybercriminals

Small and medium-sized enterprises (SMEs) are susceptible to cybersecurity vulnerabilities due to a variety of factors. Constrained financial resources frequently obstruct their capability to invest in extensive security measures, while a deficiency of proficient cybersecurity personnel intensifies these difficulties (Arroyabe et al., 2024; Rawindaran et al., 2023). Furthermore, SMEs' roles within larger supply chains make them attractive targets for cybercriminals aiming at more substantial organizations. A notable instance of this risk is the 2013 breach of Target Corporation, where attackers exploited security weaknesses in an SME contractor to access Target's systems (Plachkinova & Maurer, 2018).

The Importance of Cybersecurity for SMEs

In the contemporary landscape, where digital interactions prevail, the importance of cybersecurity is more evident. It is an essential safeguard, shielding sensitive data from malicious hazards and ensuring seamless business activities. Small and medium-sized enterprises (SMEs) must overcome this challenge like their larger counterparts by adopting robust, proactive

security measures. By doing so, they fortify their operations against potential breaches and enhance confidence among their clients and stakeholders (Taherdoost, 2024).

Cybersecurity Awareness in SMEs

Effective cybersecurity strategies prioritize risk mitigation by addressing technological infrastructures and human factors. Human error continues to represent a critical vulnerability, as cybercriminals increasingly rely on social engineering techniques to exploit individual users (Zwilling et al., 2022; Taherdoost, 2024). Consequently, organizations that neglect to cultivate comprehensive cybersecurity awareness among their personnel markedly increase their susceptibility to a comprehensive range of cyber threats (Alahmari & Duncan, 2020).

Cybersecurity awareness training equips employees with the knowledge and skills necessary to recognize and respond effectively to cyber threats. Bada and Nurse (2019) emphasize that small and medium-sized enterprises (SMEs) should prioritize cybersecurity education to compensate for their limited resources and expertise. Al-Alawi et al. (2019) highlight that such training enhances employees' willingness to implement protective measures, regardless of their roles within the organization. A security-aware workforce promotes vigilance and cooperation, substantially decreasing the risk of human errors that could result in security breaches (Taherdoost, 2024).

The National Institute of Standards and Technology (NIST) distinguishes between awareness and training, stating that awareness initiatives engage employees by emphasizing security's importance. In contrast, training ensures individuals develop specific skills to address

cyber threats. Implementing cybersecurity awareness programs enables SMEs to build resilience and reduce cyber risks.

Cybersecurity awareness and training are critical for safeguarding small and medium-sized enterprises (SMEs) from cyber threats. An informed workforce strengthens an organization's ability to prevent, detect, and respond to security incidents efficiently. By fostering a strong security culture and encouraging continuous learning, SMEs can safeguard their digital assets, conserve customer trust, and ensure long-term sustainability. Investing in cybersecurity education is not just a reactive approach but a proactive step toward securing a safer digital future (Shillair et al., 2022).

Research Gaps

Bada et al. (2019) highlighted numerous initiatives to strengthen cybersecurity resilience among small and medium-sized enterprises (SMEs) through awareness, education, and training. However, critical gaps remain in assessing their effectiveness. While these initiatives draw from academic research, industry best practices, and cross-sector collaborations, their impact remains uncertain due to a lack of empirical evaluation. Closing this research gap is essential, as SMEs—despite being the backbone of global economies—are primary targets for cyber threats owing to their constrained security resources.

A major shortcoming of existing cybersecurity education programs is their dependence on uniform, generic methods that fail to address diverse needs. These generic models often fail to effectively address the diverse and intricate cybersecurity challenges small and medium-sized

enterprises (SMEs) face. Consequently, many participants become disengaged, which diminishes the effectiveness of the training (Li et al., 2023; Hodhod et al., 2023).

SMEs encounter substantial difficulties implementing cybersecurity measures that align with their specific risk profiles without tailored solutions. This research is significant for SME owners and employees, as it aims to develop practical, customized security strategies. By doing this, it seeks to enhance their ability to safeguard their businesses against financial losses and reputational damage, ultimately strengthening their defenses in an increasingly perilous digital environment.

Furthermore, the lack of well-defined metrics for assessing cybersecurity awareness initiatives constitutes a critical research gap. Existing evaluation frameworks are either inconsistent or nonexistent, making it difficult to measure improvements in security postures and behavioral changes among SME employees (Wang et al., 2023). The absence of standardized assessment tools complicates determining these programs' long-term effectiveness and sustainability. Developing reliable evaluation frameworks benefits policymakers, industry stakeholders, and cybersecurity professionals seeking to refine training interventions for maximum impact.

Practical constraints pose significant challenges to adopting cybersecurity education within small and medium-sized enterprises (SMEs). Constrained financial and human resources, employee resistance, and the need to comply with regulatory obligations create considerable barriers to implementation. These challenges disproportionately affect SMEs, rendering them especially susceptible to a progressively intricate cyber threat environment.

Addressing these challenges necessitates research aimed at identifying effective training strategies that are scalable, adaptable, and accessible for budget-constrained businesses. The discoveries acquired from such research will benefit SMEs, cybersecurity trainers, and regulatory bodies by fostering practical and cost-effective security solutions.

Recent research underscores the vital importance of security awareness training in enhancing the cybersecurity framework of small and medium-sized enterprises (SMEs). However, deficiencies in its execution and assessment impede its overall efficacy. Many small business owners rely on such training to mitigate cyber risks but often lack the structured guidance to design and execute effective programs. Tackling these research gaps is essential for developing cybersecurity strategies that provide substantial, long-term benefits while strengthening SMEs against increasingly sophisticated cyber threats. By improving SMEs' capacity to defend against cyberattacks, this research ultimately fosters economic stability, ensures business continuity, and supports the integrity of the broader digital ecosystem.

Research Questions and Methodology

Research Questions

RQ 1: What strategies and measures does K2 Partnering Solutions employ to mitigate and counter potential cyber threats to its systems and data?

Context: This research question aims to analyze and understand the organization's overall cybersecurity measures, identify gaps, and provide recommendations for enhancing defenses against threats in the digital landscape.

RQ 2: What strategies does K2 Partnering Solutions implement to raise employee awareness and provide training to protect the business from potential cyberattacks?

Context: This research question examines the organization's methods for improving cyber risk awareness and training among its employees. It aims to understand how management communicates potential cyber threats and trains staff to handle them effectively, fostering a cybersecurity awareness culture.

Research Methodology

This study will employ a qualitative research design to conduct an in-depth case analysis of K2 Partnering Solutions' approach to cybersecurity. Qualitative methods allow researchers to closely observe, understand, and investigate various phenomena. This approach facilitates the collection of rich insights and fosters a profound comprehension of the intricate interactions of human behavior and experiences (Goldberg & Allen, 2015).

The case study will employ a purposive sampling strategy to examine past and present cyber threats encountered by K2 Partnering Solutions. The analysis will investigate the relationship between these threats and several factors, including insufficient cybersecurity protocols, a complacent approach to protecting sensitive information, and constrained financial resources.

Data will be collected from ten key stakeholders through semi-structured face-to-face interviews, online questionnaires, and a review of company documents. The data triangulation method, a qualitative research approach, will be used for data analysis. This method aims to

cross-verify findings and enhance conclusions by incorporating diverse perspectives, providing a holistic understanding of the organization's cybersecurity posture.

SECTION 2: Literature Support

Description of the Problem and Business Practices

The rapid digital transformation has unlocked significant opportunities for growth, efficiency, and connectivity for small and medium-sized enterprises (SMEs). However, it has also heightened their exposure to cybersecurity threats, with inadequate education and awareness of data security practices as a significant contributing factor (Zwilling et al., 2022; Chidukwani et al., 2022; Bada et al., 2019).

The 2023 Verizon Data Breach Investigations Report indicates a concerning statistic: almost 43% of cyberattacks explicitly target small and medium-sized organizations (SMEs), highlighting their increasing vulnerability in the swiftly changing digital environment. This worrisome trend is further compounded by Rawindaran et al. (2023), Comerford (2022), and Faulhaber-Brad Moon (2023), who highlight that, unlike larger corporations with devoted Information Technology teams and advanced security infrastructures, SMEs often operate with constrained resources and insufficient cybersecurity expertise, leaving them at a significant disadvantage.

Consequently, small and medium-sized enterprises (SMEs) become prime targets for cybercriminals, a concern highlighted by Olney (2023). This lack of preparedness exposes vulnerabilities that can be exploited, allowing unauthorized access to delicate business information. As cyber threats grow in sophistication, a reactive attitude to cybersecurity no longer suffices. This argument is supported by Luna (2025), Chidukwani et al. (2022), and Prümmer et al. (2024), who assert that SMEs must shift towards a proactive cybersecurity

strategy—one that emphasizes continuous education and robust defense mechanisms to mitigate risks effectively. Without consistent and well-planned investment in cybersecurity, SMEs continue to be at risk of cyberattacks that could disrupt their operations and jeopardize their long-term viability in a constantly shifting digital landscape.

Critical Review of Prior Research

Researchers, including Prümmer et al. (2024), Nurse (2021), and Papathanasiou et al. (2024), underscore an important concern: many owners and employees of small and medium-sized businesses lack awareness of essential cybersecurity practices. These practices include implementing strong password policies, recognizing phishing attempts, and ensuring timely software updates. This knowledge gap significantly heightens their risk of data breaches, resulting in financial losses, reputational damage, operational disruptions, and potential legal repercussions.

Tolossa (2023), Olabode (2023), and Bada et al. (2019) emphasize the precarious need for customized cybersecurity awareness campaigns, government-led initiatives, and cost-effective security strategies bespoke to the explicit struggles encountered by small and medium-sized enterprises (SMEs). Despite widespread recognition of these requirements, many SMEs struggle with establishing comprehensive cybersecurity safeguards due to cost constraints, time constraints, and the perceived complexity of cybersecurity policies (Shillair et al., 2022; Junior et al., 2023). To effectively address these susceptibilities and strengthen their defenses, SMEs must weave cybersecurity awareness into their corporate culture. This strategy will foster a proactive

security mindset and strengthen the organization's resilience in the constantly shifting cyber threat landscape.

This study seeks to tackle the significant gap in cybersecurity awareness within SMEs by investigating its root causes, assessing the impact of current training and educational efforts, and proposing actionable strategies to improve security without imposing undue burdens. Drawing on discoveries from established literature and practical case analyses, this study offers small and medium-sized enterprises (SMEs) a practical and strategic framework for enhancing their cybersecurity resilience, thereby mitigating the threat of data breaches and cyberattacks that could jeopardize their long-term sustainability.

Cybersecurity Challenges in SMEs

Cybersecurity, as defined by the United States Cybersecurity and Infrastructure Security Agency, is "the art of protecting networks, devices, and data from unauthorized access or criminal use while ensuring the confidentiality, integrity, and availability of information." This concept, as articulated by Senol and Karacuha (2020), represents a dynamic and evolving process aimed at safeguarding electronic systems from malicious attacks. Given the rapid and constant emergence of new threats and countermeasures, cybersecurity is an ongoing challenge that requires continual adaptation and vigilance.

Chidukwani et al. (2022). Arroyabe et al. (2024) and Junior et al. (2023) highlighted that small and medium-sized enterprises (SMEs), which represented 90% of businesses globally and employed over half of the world's workforce, were increasingly dependent on advancements in information technology (IT) and broadband connectivity. These technological advancements

have significantly transformed SME operations by improving efficiency, streamlining processes, and enhancing customer interactions. However, this digital transformation also introduced new risks that SMEs had to navigate.

Bada and Nurse (2019), Mugwagwa et al. (2024), and Taherdoost (2024) highlighted that while IT and broadband infrastructure provide SMEs with competitive advantages—such as automation, real-time communication, and data analytics—they also present significant vulnerabilities. The integration of digital technologies exposes SMEs to a constantly evolving cyber threat landscape, making it essential for them to adopt a strategic methodology for cybersecurity. This strategy is necessary for optimizing digitalization's advantages while mitigating risks.

Arroyabe et al. (2024) projected that the worldwide financial impact of cybercrime will rise from $9.5 trillion in 2024 to $10.5 trillion in 2025, indicating a significant enhancement in the intricacy and prevalence of cyber threats. This emerging trend highlights the imperative for improved cybersecurity protocols, particularly in large organizations versus SMEs. Large firms possessed the capacity to establish extensive security frameworks, whereas SMEs encountered disadvantages, bearing disproportionate risks due to their constrained cybersecurity infrastructure. Mugwagwa et al. (2024) contended that while SMEs exhibited greater agility in IT management than larger firms, their financial and technical limitations rendered them especially susceptible to cyberattacks. The economic impact of such disasters may be much worse for SMEs, highlighting the necessity for them to embrace proactive cybersecurity methods to minimize disruptions.

Tetteh (2024) argued that small and medium-sized enterprises (SMEs) faced a remarkable challenge due to their inadequate resources and deficiency of specialized cybersecurity expertise. He contended that many of these businesses relied on outdated software and hardware, which created security vulnerabilities easily exploited by cybercriminals. Furthermore, Tetteh pointed out that financial constraints often prevented SMEs from investing in comprehensive cybersecurity measures or hiring dedicated IT professionals. As a result, he asserted, this lack of investment heightened their vulnerability to cyber threats, especially when compared to larger organizations that benefited from more advanced and well-resourced security infrastructures.

Moreover, Bada and Nurse (2019) argued that insufficient cybersecurity awareness among personnel was an acute factor contributing to the vulnerability of SMEs. As Tetteh (2024) observed, cybercriminals regularly exploit human error through phishing, social engineering, and other tactics. Without proper training, employees were more likely to inadvertently provide cybercriminals access to sensitive company data, thus raising the risk of a breach. In line with this, Mugwagwa et al. (2024) noted that SMEs typically had weaker technological defenses and lower cybersecurity awareness, rendering them especially appealing targets for cybercriminals.

Chidukwani et al. (2022) further emphasized that numerous SMEs neglected to implement satisfactory cybersecurity measures despite their substantial role in national economies, rendering them susceptible to evolving cyberattacks. Consequently, the growing threats to SMEs emphasize the pressing need for proactive cybersecurity protocols, enhanced employee training, and thorough risk management to protect their operations and sensitive information.

Why Cybercriminals Target SMEs

Arroyabe et al. (2024) and Rüdell (2024) observed that cybercriminals are progressively targeting small and medium-sized enterprises (SMEs) due to their inherent vulnerabilities. The authors argued that SMEs typically lack the financial and technological resources to institute robust cybersecurity measures. Rüdell (2024) contended that numerous SMEs had an erroneous notion that they are less susceptible to cyberattacks than larger firms, resulting in complacency in their cybersecurity measures. This combination of limited resources and misplaced confidence renders SMEs prime targets for cyber threats, including economic exploitation and data breaches. Recognizing these vulnerabilities is essential for formulating effective cybersecurity solutions for SMEs in the current digital environment.

False Sense of Security and Lack of Cybersecurity Investment

Arroyabe et al. (2024) and Junior et al. (2023) argued that cybercriminals often target small businesses due to a prevalent misconception among small and medium-sized enterprise (SME) owners that their organizations are less likely to be attractive targets. Many SME owners operate under the false belief that their smaller size or limited visibility provides a shield against cybercriminals. This misunderstanding significantly underestimates their digital threats, resulting in insufficient investment in robust cybersecurity measures. As a result, SMEs become particularly vulnerable to cyberattacks, with criminals efficiently exploiting these weaknesses. Therefore, the false sense of security among SMEs directly contributes to their heightened exposure to cybercrime risks.

Limited Resources and Vulnerability to Sophisticated Attacks

Chidukwani et al. (2022) and Chaudhary et al. (2023) highlighted the significant difficulties encountered by small and medium-sized organizations (SMEs), primarily due to their limited financial and technical resources. This shortage of resources starkly contrasted with the more robust cybersecurity measures employed by larger organizations, which had the means to implement sophisticated, multi-layered security protocols. In contrast, SMEs often relied on essential, consumer-grade security tools, which, while cost-effective, were insufficient to defend against increasingly complex cyberattacks targeting more financially lucrative entities. Tetteh (2024) further supported this view, emphasizing that the resource disparity left SMEs particularly susceptible to threats such as ransomware, phishing, and data breaches. Consequently, while larger organizations could invest in sophisticated security measures to prevent these risks, SMEs often struggled to secure even basic protections, making them prime targets for cybercriminals.

SMEs as Gateways to Larger Organizations

Mugwagwa et al. (2024) underscore a critical vulnerability encountered by small and medium-sized enterprises (SMEs), rendering them appealing targets for cybercriminals: their function as intermediaries for larger organizations. This perspective is further validated by Wong et al. (2022) and Sharma (2025), who observe that SMEs face heightened cyber risks within global supply chains, mainly due to frequently inadequate cybersecurity measures. Moreover, Teymourlouei (2018) and Krahl (2019) point out that the limited resources and deficient security frameworks common among SMEs create an environment conducive to cybercriminal activities.

A cyber breach in an SME can result in extensive financial and operational disruptions, impacting entire industries. Consequently, strengthening the cybersecurity of SMEs is essential to safeguard individual businesses and maintain the stability and integrity of the global economy.

Exploitation of Weaknesses in Third-Party Vendors

Tetteh (2024) highlighted the significant vulnerability of SMEs to cyberattacks due to their reliance on third-party vendors, such as managed service providers (MSPs), for various business operations. Cybercriminals often exploit weaknesses in these third-party relationships to infiltrate multiple organizations simultaneously. Tetteh (2024) cited research indicating that four out of five organizations had experienced security breaches linked to vulnerabilities in third-party vendors. As SMEs often operate within more extensive supply networks, a single cybersecurity lapse within one organization could lead to widespread exposure to cyber risks across multiple entities. Moreover, Tetteh underscored the growing prevalence of automated cyberattacks and the vulnerabilities inherent in supply chains, which further heightened the exposure of SMEs to cybercriminal activities.

Vulnerabilities in Sensitive Sectors: The Healthcare Example

Javaid et al. (2023) emphasized the heightened vulnerability of SMEs to cybercriminals, particularly within sensitive sectors such as healthcare. They highlighted that the healthcare sector's reliance on a broad network of suppliers and service providers made it especially susceptible to cyberattacks. Small businesses operating within the healthcare ecosystem frequently exchange sensitive data with larger healthcare organizations, creating numerous points of entry for cybercriminals. A breach at a small business could expose critical patient

information, underscoring the interconnected nature of cybersecurity risks. Javaid et al. (2023) stressed that safeguarding the security of small businesses in these sectors was crucial in preventing widespread data breaches and maintaining public trust in essential healthcare services.

Cybersecurity Awareness and Education Gaps in SMEs

Cybersecurity knowledge is crucial for protecting organizational digital assets; nonetheless, notable deficiencies persist in the current literature, especially regarding small and medium-sized organizations (SMEs). These organizations are especially susceptible to cyber risks due to constrained resources and a deficiency of comprehensive training programs (Chidukwani et al., 2022; Arroyabe et al., 2024). Shaw et al. (2009) defined cybersecurity awareness as recognizing the importance of information security and personal accountability in maintaining security measures. However, recent research, such as Abrahams et al. (2024), accentuated the increasing prominence of human factors in managing cyber risks. The authors noted that while technological solutions like firewalls, encryption, and intrusion detection systems are essential, they are insufficient. The real challenge lies in the lack of a well-rounded, integrated educational approach within many SMEs that addresses technical knowledge and fosters behavioral change. This gap accentuates the requirement for a more holistic cybersecurity education strategy combining technical and human-focused approaches.

The swift evolution of technology and the expansion of the digital landscape have been extensively explored by researchers such as Zwilling et al. (2022), Arroyabe et al. (2024), and Nurse (2021), whose findings highlight the increased susceptibility to cyber threats resulting

from these developments. The authors observed that the increase in mobile devices and the Internet of Things (IoT) has further exacerbated vulnerabilities, particularly within small and medium-sized enterprises (SMEs), which often struggle to engage employees effectively through comprehensive awareness campaigns. Conventional training methods, including compulsory sessions and static reading materials, are insufficient in promoting lasting behavioral changes, as noted by Nurse (2021). There is a substantial gap in the literature for more dynamic and interactive approaches to cybersecurity education, moving beyond basic technical training. As Nurse (2021) proposed, the integration of gamification and simulations presents a promising avenue for addressing this gap, as these methods can enhance employee engagement while reinforcing key security practices.

Shillair et al. (2022) and Chaudhary et al. (2023) accentuated the persistent and troubling deficiencies in cybersecurity awareness, specifically within small and medium-sized enterprises (SMEs), which remain woefully unprepared for the changing landscape of cyber threats. This vulnerability is motivated by a confluence of causes, including insufficient resources, a deficiency of cybersecurity expertise, and a widespread underestimation of the severity of cyber risks.

Moreover, SMEs' exposure to cybercriminal activity is exacerbated by human factors, such as poor password management and an inability to recognize phishing attempts (Javaid et al., 2023). This reveals a glaring gap in the current literature: While human error is universally acknowledged as a leading cause of cybersecurity breaches, there is a striking lack of targeted, effective training solutions explicitly designed to address and mitigate these errors. Research by Taherdoost (2024) and Chaudhary et al. (2023) emphasized the pivotal role of human error in

security incidents, underscoring the need for specialized, context-driven educational programs that go beyond general awareness to directly address the underlying causes of human-induced security vulnerabilities.

The existing literature reveals a notable and concerning gap: the absence of comprehensive guidance on effectively integrating leadership endorsement into cybersecurity awareness programs. Abrahams et al. (2024) emphasized that leadership is essential for advancing cybersecurity best practices and maintaining employee accountability in protecting the organization's digital assets. Despite this acknowledged significance, there remains a lack of research offering actionable strategies for incorporating leadership into cybersecurity training initiatives. This gap is particularly alarming, as leadership endorsement is crucial for the effective execution of cybersecurity policies and for cultivating an organizational culture that prioritizes security. Without well-defined frameworks for leadership involvement, organizations risk undermining the foundation of their cybersecurity efforts and failing to cultivate a culture of collective accountability and commitment to security at all levels.

An evident deficiency in the current literature highlights the pressing necessity for more adaptable and personalized cybersecurity training methodologies. Conventional training programs often rely on stale content and fail to address cyber threats' dynamic and continually evolving nature. As Abrahams et al. (2024) and Prümmer et al. (2024) noted, it is crucial to create adaptive training programs that change in response to emerging threats and trends. However, despite these recommendations, there is a conspicuous absence of comprehensive frameworks for creating such adaptive systems, particularly those that provide real-time feedback and personalized learning paths. While models like Taherdoost's (2024) Integrated

Cybersecurity Awareness Training (iCAT), which integrates gamification and knowledge graphs, show considerable promise, the existing literature remains limited regarding the effective implementation and evaluation of these innovative approaches, especially within resource-constrained small and medium-sized enterprises (SMEs). This gap accentuates the pressing necessity for further research to investigate how adaptive, personalized training models can be scaled and optimized for diverse organizational contexts. Ensuring these models adapt to cyber threats and promoting meaningful and lasting behavioral change is essential.

Small and medium-sized enterprises (SMEs) face significant obstacles in effectively measuring the impact of their cybersecurity awareness programs (Arroyabe et al., 2024), a gap that remains underexplored in the literature. Prümmer et al. (2024) observed that the absence of reliable metrics hinders organizations from effectively evaluating the success of their training initiatives and implementing data-driven adjustments. Taherdoost (2024) accentuated the imperative for real-time tracking and feedback mechanisms to drive continuous improvement by providing insights into employee engagement and program effectiveness. The literature, however, lacks practical and concrete evaluation frameworks specifically designed to address the resource limitations of SMEs. Future research should focus on developing adaptable, interactive training models that incorporate real-world simulations, continuous feedback, and engaging components to drive lasting behavioral change among employees.

Additionally, the integration of leadership involvement in cybersecurity training, as advocated by Shaw (2024) and Simmons and Sundaresan (2024), remains underexamined and warrants further exploration. Establishing clear guidelines for leaders is essential to enhance cybersecurity awareness and embed security within organizational culture. Finally, there is a

critical need for research into clear, actionable metrics for evaluating the efficacy of training programs, enabling SMEs to refine their approaches and remain proactive against emerging cyber threats. Addressing these gaps will enhance the ability of SMEs to implement practical cybersecurity training and better protect their digital assets.

Common Cybersecurity Threats and Vulnerabilities

Nurse (2021) posited that cyber-attacks could manifest in various forms, targeting a broad spectrum of vulnerabilities. Arroyabe et al. (2024) and Javaid et al. (2023) further emphasized the significant threats posed by cybersecurity vulnerabilities, highlighting the serious risks they presented to data integrity, operational continuity, and financial stability. The authors contended that external threats and internal weaknesses required vigilant attention from organizational leaders. They outlined the most prevalent cybersecurity threats and internal vulnerabilities that could substantially impair an organization's security posture, emphasizing the urgent need for holistic strategies to mitigate these hazards effectively.

Malware

Malware, which encompasses viruses, worms, trojan horses, ransomware, and spyware, is explicitly designed to damage, interfere with, or unlawfully access computer systems. Arroyabe et al. (2024) asserted that malware directly threatens the integrity and functionality of technological infrastructure. As Javaid et al. (2023) point out, a single malicious link can suffice to introduce malware into a network, highlighting the ease with which these attacks can proliferate.

Phishing

Phishing attacks deceive individuals into disclosing confidential personal information, including passwords, credit card numbers, and account login details. Phishing attacks, often conducted via deceptive emails, texts, or websites that imitate authentic sources, exploit human trust, rendering them highly successful in compromising security (Arroyabe et al., 2024).

Denial-of-Service (DoS) and Distributed Denial-of-Service (DDoS) Attacks

Denial of Service (DoS) and Distributed Denial of Service (DDoS) attacks inundate system resources, incapacitating a network or website for authorized users. The DDoS variant is particularly concerning, as it harnesses multiple compromised devices, complicating efforts to mitigate the attack and restore normal operations (Arroyabe et al., 2024).

Man-in-the-Middle (MitM) Assaults

MitM assaults transpire when an assailant intercepts communication between two entities, frequently without the awareness of either party. This assault may entail intercepting unencrypted Wi-Fi communications or altering data between a user and a website, potentially leading to the theft or modification of sensitive information.

SQL Injection

SQL injection uses weaknesses in website databases, enabling attackers to insert harmful SQL commands. If successful, attackers can change, extract, or eliminate vital data, jeopardizing the website's integrity and related information (Arroyabe et al., 2024).

Cross-Site Scripting (XSS)

XSS attacks entail the injection of malicious scripts into a website, which are subsequently executed in the browsers of visitors accessing the site. These scripts can steal sensitive data or execute unauthorized operations on behalf of the user, undermining user privacy and security (Arroyabe et al., 2024).

Ransomware

Ransomware encodes files on an infected system, making them inaccessible, and demands a ransom—typically paid in cryptocurrency—for the decryption key. The repercussions of ransomware can be catastrophic, especially for firms dependent on uninterrupted access to essential data (Arroyabe et al., 2024).

Zero-Day Exploits

Zero-day attacks take advantage of previously unidentified weaknesses in software or hardware, which remain unaddressed by the vendor until a corrective update is deployed. The inherent danger of these attacks lies in their ability to leverage unresolved security flaws before any defensive measures can be enacted (Arroyabe et al., 2024).

Internal Vulnerabilities within Organizations

Arroyabe et al. (2024) emphasized that organizations must address external threats and confront several internal vulnerabilities that pose significant risks. These vulnerabilities originate from flaws in a company's internal systems, processes, or practices, rendering them susceptible

to exploitation by both external attackers and insiders. The key internal vulnerabilities identified by Arroyabe et al. (2024) and Javaid et al. (2023) include:

Weak Passwords

Vulnerable and easily predictable passwords constitute a major cybersecurity weakness, granting attackers straightforward access to sensitive networks. The 2020 Verizon Data Breach Investigations Report indicates that 81% of breaches associated with hacking involve compromised passwords, underscoring the necessity for enhanced authentication protocols. Javaid et al. (2023) also found that inadequate passwords are a principal target for cybercriminals, advocating for enterprises to enforce stringent password regulations. Employing intricate and distinctive passwords and supplementary security protocols can diminish the likelihood of assaults and safeguard essential corporate information.

Insufficient Access Controls

Inadequate access control mechanisms allow employees to access systems and data beyond their designated roles. Such over-privileged access heightens the risk of data breaches and misuse, especially if unapproved persons exploit these privileges.

Outdated Software and Systems

Neglecting to update and patch software and systems consistently renders organizations susceptible to recognized vulnerabilities. Maintaining systems with the latest updates is crucial for safeguarding against emerging threats and preventing vulnerabilities from being exploited.

Lack of Security Training

Employees lacking cybersecurity training are more exposed to social engineering attacks and manipulation. Javaid et al. (2023) emphasize that since cybercriminals persist in innovation, ongoing and thorough employee training in cybersecurity best practices is essential for sustaining organizational resilience.

Insider Threats

Employees can significantly threaten an organization's security, whether malicious or negligent. Insider threats from deliberate sabotage or ignorance of security protocols can lead to serious data breaches and operational disruptions.

Inadequate Network Security

Weak network security, particularly unsecured wireless networks, can allow attackers to infiltrate internal systems. Implementing robust encryption and authentication measures is critical to prevent unauthorized access and safeguard organizational assets.

Unchecked Usage of Removable Media

Malware and data breaches are more likely when removable media, such as USB drives, are not regulated. To lessen these dangers, businesses should implement rules restricting the usage of these gadgets.

Insecure Configuration Settings

Default configurations left unchanged on hardware, software, or network devices can expose systems to potential susceptibilities. Ensuring that devices are securely configured is critical to reducing the attack surface and preventing easy exploitation by cybercriminals.

Inadequate Incident Response Planning

Without a well-articulated incident response strategy, companies may lack the readiness to identify, control, and alleviate security breaches. A robust incident response strategy is crucial to mitigate damage and facilitate a rapid recovery in the occurrence of an attack.

Poor Physical Security

Unauthorized personnel may be able to access vital infrastructure due to inadequate physical security measures. Combining cybersecurity and physical security measures can avoid intrusions from illegal physical entry.

Data Storage and Transmission Insecurity

Neglecting to implement encryption when storing or transmitting sensitive information significantly elevates the likelihood of illegal access. Strong encryption protocols are essential for ensuring the protection of data both during transmission and while stored.

Vendor and Third-Party Risks

Dependence on third-party services or products without comprehensive security assessment can bring substantial risks into an organization's ecosystem. Ensuring a safe environment is crucial for verifying that all third-party providers comply with stringent security criteria.

The growing incidence of data breaches underscored the evolving tactics employed by cybercriminals and the expanding attack surface of contemporary IT environments. Appel et al. (2023) discovered that over 70% of global data breaches were committed by external actors, with nearly half involving unauthorized access or intrusion. While financial gain drove approximately 86% of these breaches, the Cost of a Data Breach Report 2020 (IBM, 2020) highlighted that cyber incidents were also motivated by various factors, including ideology, personal grievances, espionage, state-sponsored activities, human error, and even amusement. This aligned with the findings of Chidukwani et al. (2022), who identified web application servers as the most frequently targeted IT assets, primarily due to the increasing reliance on web-based applications and cloud-based software-as-a-service (SaaS) platforms. In addition to web application servers, other critical targets included end users, email servers, database servers, and user endpoints such as desktops and laptops. Moreover, as highlighted by Arroyabe et al. (2024), mobile and Internet of Things (IoT) devices were particularly vulnerable in small and medium-sized business (SME) environments, often serving as key entry points for cyber attackers. Together, these findings emphasized the intricate nature of cyber threats, where external adversaries exploited various assets and motivations, particularly within an increasingly digital and interconnected landscape.

Based on Verizon's 2024 Data Breach Investigations Report (DBIR), cyber risks have risen sharply, with documented breaches almost doubling from the prior year to 10,626 events. The report identifies stolen credentials as the initial attack vector in 24% of breaches, emphasizing the urgent need for more vigorous password hygiene and multifactor authentication (MFA). Phishing attacks accounted for 14% of credential theft incidents, with users frequently falling for phishing emails in under 60 seconds. Ransomware remained a pervasive threat, comprising 70% of malware incidents and affecting 92% of industries, including financial services, healthcare, and public administration.

Moreover, according to the same report, human error was a major contributor to breaches, responsible for 28% of incidents, highlighting the necessity of continuous employee training. Additionally, 15% of breaches stemmed from third-party vulnerabilities, underscoring the significance of supply chain security and vendor risk management. The financial impact of cyberattacks, particularly ransomware, was substantial, with median ransom demands amounting to 1.34% of a victim’s revenue and median adjusted losses reaching $46,000, according to Verizon’s 2024 DBIR. A critical concern in password management was also noted, with 74% of passwords found in SpyCloud’s database being reused, creating significant security risks. These discoveries reinforce the pressing need for enhanced cybersecurity measures, including stronger authentication practices, employee education, and improved third-party risk management (Hylender et al., 2024).

Impact of Cyberattacks on SMEs

Financial Losses

Cyberattacks present a substantial financial and operational risk to small and medium-sized enterprises (SMEs), with possible expenses varying from $25,000 to $3 million (Kindle, 2025). The financial repercussions are significant; Clearly (2023) and Woodgate (2024) estimated that the average expense of a cyberattack on SMEs was roughly $200,000 and $250,000, respectively—sums that may be catastrophic for resource-constrained enterprises.

Moreover, the enduring ramifications of cyber disasters surpass the immediate financial losses. According to Galvin (2018), nearly 60% of small businesses fail within six months following a cyberattack. In addition to financial damage, cyberattacks disrupted essential operations, undermined customer trust, and jeopardized business continuity. As noted by Chidukwani et al. (2022), these findings accentuate the severe vulnerability of SMEs in the ever-evolving cyber threat landscape and highlight the pressing need for comprehensive cybersecurity strategies to ensure their survival and resilience.

Kindle (2025) highlighted the severe and multifaceted financial consequences of cyberattacks, positioning them as one of the most critical threats to business stability. Cyberattacks lead to immediate revenue losses due to operational disruptions and decreased productivity, while employees are locked out of essential systems, further amplifying financial strain. Ransomware attacks, in particular, force businesses to pay exorbitant ransom without data recovery guarantees, adding to the financial burden. Additionally, legal costs can escalate into the millions, especially in cases involving regulatory violations or class-action lawsuits. Beyond

these direct costs, businesses must invest heavily in cybersecurity, hiring experts to mitigate risks and address vulnerabilities. Kindle (2025) emphasized the necessity of transitioning from a reactive to a proactive cybersecurity strategy to protect financial stability and guarantee long-term resilience in a progressively intricate threat landscape.

Damage to Customer Business Reputation

Cyberattacks can profoundly impact a business's reputation by undermining customer trust, which is crucial for long-term success. Customers expect companies to safeguard their data, and when a breach occurs, confidence can plummet, leading to decreased sales and diminished brand credibility. The 2024 Hiscox Cyber Readiness Report (Hiscox, 2025) revealed that 47% of organizations that faced a cyberattack in 2023 found it difficult to attract new customers, while 43% reported losing existing clients. This loss of trust may result in enduring reputational harm.

Furthermore, the 2024 Okta for Good Impact Report indicates that over 40% of small and medium-sized businesses (SMEs) experienced a decline in customer confidence following a cyberattack, with nearly 40% acknowledging significant damage to their reputation (Okta, 2024). According to the same report, the repercussions are even more severe for larger SMEs, where 50% noted substantial adverse effects on trust and reputation. In some instances, companies may need to invest in marketing and public relations strategies, such as offering discounts or promotions, to rebuild their image and win back customers. These insights highlight that cyberattacks diminish customer trust and restrict business opportunities, ultimately threatening customer loyalty and the organization's long-term viability.

Barriers to Implementing Cybersecurity Measures

Small and medium-sized enterprises (SMEs) are particularly vulnerable to cyberattacks due to inadequate resources, as highlighted by Papathanasiou et al. (2024), Heidt et al. (2019), and Chidukwani et al. (2022). Budget limitations often prevent SMEs from allocating funds for comprehensive IT security measures, and managing directors, IT staff, and service providers recognize the difficulty of prioritizing security initiatives with such limited resources. Unlike larger organizations with dedicated cybersecurity teams and advanced frameworks, SMEs face a constantly evolving landscape of cyber threats. This disparity increases their exposure to attacks and severely constrains their capability to react to and recover from breaches. These findings emphasize that targeted investments in cybersecurity are crucial for strengthening the defenses of smaller businesses and ensuring their long-term resilience.

Complexity of Solutions

Implementing cybersecurity solutions in small and medium-sized enterprises (SMEs) poses substantial financial, informational, and structural constraints. Chidukwani et al. (2024) found that, in addition to financial limitations, a key obstacle was the lack of cybersecurity awareness and expertise within these organizations. Numerous SMEs struggle to comprehend and adopt established cybersecurity frameworks, such as ISO 27001, which offer industry-recognized best practices for securing IT systems (Rawindaran et al., 2023). This knowledge deficiency not only undermines their capacity to counter cyberattacks but also impedes their ability to comply with regulatory mandates, rendering them more susceptible to security breaches.

Similarly, Auyporn et al. (2023) ascribed the low adoption rate of cybersecurity standards among SMEs to constrained financial and technical resources. Cybersecurity frameworks are essential for standardizing risk management and sharing knowledge (Cisternelli, 2024). However, despite their strategic benefits, SMEs often lack the financial and human capital necessary for effective implementation (Tetteh, 2024; Chidukwani et al., 2022). The intersection of financial strain, limited awareness, and technical shortcomings creates a significant gap in cybersecurity resilience. Addressing this disparity necessitates a multifaceted strategy encompassing focused training programs, financial support, and the creation of accessible cybersecurity solutions designed to meet the specific issues encountered by SMEs. Small firms would persistently encounter increased risks without these actions, underscoring the pressing necessity for organized support networks to bolster cybersecurity adoption and resilience in this vital sector.

Perceived Low Return on Investment (ROI)

A noteworthy obstacle to advancing the adoption of appropriate cybersecurity measures in small and medium-sized enterprises (SMEs) is the perceived minimal return on investment (ROI). This view mainly stems from an insufficient understanding of prospective risks and the enduring ramifications of cyberattacks. Chidukwani et al. (2024) noted that some SME proprietors fail to acknowledge the vital significance of investing in cybersecurity, frequently perceiving it as an extraneous cost rather than a fundamental business requirement. This mindset is common among those unaware of the enduring cyber-attack hazards, such as data breaches, reputational harm, and significant financial losses. As a result, the immediate financial expenditures associated with cybersecurity are frequently assessed against the perceived

negligible short-term advantages, perhaps resulting in postponements in implementation or insufficient protective measures.

Heidt et al. (2019) further highlighted that the lack of awareness regarding the potential severity of cyberattacks underscores the pressing need for improved education and understanding of cybersecurity's long-term value. Fostering this awareness is crucial for encouraging proactive investments in protective measures, ensuring that SMEs can effectively safeguard their operations and mitigate risks.

Lack of Incident Response Planning

The Cybersecurity and Infrastructure Security Agency (CISA, 2025) asserts that cyber incident response is crucial for enterprises to minimize harm from cyber incidents, including data breaches and service interruptions. An efficient incident response plan (IRP) ensures rapid recovery of operations while safeguarding assets and reputation. Such a plan should clearly delineate roles, responsibilities, and communication protocols, enabling staff to respond promptly and effectively to mitigate the impact of security incidents that could cause significant operational and reputational damage.

Nicholls (2024) found that small and medium-sized enterprises (SMEs) face challenges in developing and implementing effective incident response policies due to limited resources and expertise. Nonetheless, SMEs need to adopt a structured six-step approach for incident response: preparation, detection, containment, elimination, recovery, and post-incident analysis. By consistently evaluating and enhancing their strategies, SMEs can mitigate cyber-attack effects, maintaining operational continuity and resilience against advancing cyber threats.

Cybersecurity Frameworks and Standards

Taherdoost (2022) contended that cybersecurity standards provide clear and structured guidelines that enable organizations to define and achieve their cybersecurity goals while reducing the likelihood of cybercrimes. These standards offer proven practices and assessment methods to guarantee information systems meet established security requirements. Chidukwani et al. (2022) emphasized that cybersecurity frameworks tailored for small and medium-sized enterprises (SMEs) enhance their ability to strengthen information security and manage cybersecurity risks effectively. These frameworks assist SMEs in recognizing and addressing cyber threats, enhancing communication regarding security matters, standardizing the provision of secure services, and boosting overall operational efficiency.

Both perspectives compellingly highlight the critical importance of structured cybersecurity practices in safeguarding organizational resilience. Taherdoost (2022) argued that cybersecurity standards are crucial for preventing cyberattacks, reducing risks, and ensuring operational continuity. In contrast, Chidukwani et al. (2022) emphasized the usefulness of cybersecurity frameworks in helping small and medium-sized enterprises (SMEs) align their practices with established best practices, thereby enhancing operational effectiveness. Additionally, Taherdoost noted that standardized approaches facilitate regulatory compliance and provide a consistent basis for evaluating and comparing security systems across international borders. This capacity for international comparison greatly promotes the broad adoption of cybersecurity standards, strengthening joint efforts to safeguard digital assets from a progressively intricate threat environment.

Cybersecurity standards are typically divided into information security and governance requirements. The International Organization for Standardization (ISO) 27000 series, Information Security Forum Standard of Good Practice (ISF SOGP), National Institute of Standards and Technology (NIST) 800 series, Sarbanes-Oxley Act (SOX), and Risk IT Framework are examples of information security standards that specifically address security challenges. The requirements of an organization determine which standard or framework is best suited. Chidukwani et al. (2024) argued that the NIST Cybersecurity Framework (NIST CSF) is especially favored by small and medium-sized enterprises (SMEs) because of its standardized methodology for controlling cybersecurity risks. Nonetheless, this highlights a crucial issue: Although the NIST CSF provides a systematic framework, it is necessary to recognize that it may not fully cater to the distinct requirements of SMEs. Consequently, SMEs must meticulously assess which framework or standard most effectively corresponds with their cybersecurity needs, ensuring it transcends a generic solution to tackle their threats.

Al-Dosari and Fetais (2023) contended that although established frameworks such as NIST offer significant guidance, they may be excessively theoretical or inflexible for small and medium-sized organizations (SMEs). Their investigation indicated that these frameworks' severe requirements and inflexibility hinder their practical implementation in smaller firms, which frequently encounter quickly evolving technical environments. The rising popularity of cloud computing and platform-based services has intensified this issue, further heightening the vulnerability of SMEs. Al-Dosari and Fetais (2023) argued that IT security standards must evolve to be more dynamic and responsive to evolving threats to handle these difficulties. Considering these constraints, SMEs ought to embrace a more adaptable strategy to

cybersecurity, integrating diverse frameworks as necessary to mitigate their risks and security difficulties successfully. This adaptability would allow SMEs to remain proactive against emerging risks without being limited by inflexible, uniform solutions.

Security Information and Event Management (SIEM) Systems

Manzoor et al. (2024) underscored the urgent cybersecurity challenges small and medium-sized enterprises (SMEs) face. They noted their increased exposure to cyber threats from resource constraints, including constrained budgets, insufficient personnel, and a deficiency in specialized technical skills. These constraints hinder SMEs from establishing robust security processes, rendering them vulnerable to increasingly sophisticated attacks. The authors proposed adopting scaled-down Security Information and Event Management (SIEM) systems as a cost-effective solution to address these challenges. These solutions facilitate critical real-time surveillance and allow prompt reactions to security problems, thereby improving the cybersecurity stance of SMEs while reducing financial pressures.

While traditional commercial SIEM systems offer a broad range of functionalities, their high costs often render them prohibitive for many SMEs. Manzoor et al. (2024) advocated for open-source and cloud-based SIEM systems as a more economical alternative, delivering critical features such as real-time threat detection and customization at a significantly reduced cost. However, according to Manzoor et al. (2024), implementing these solutions presents challenges. Open-source options require higher technical proficiency and may lack the additional functionalities and support typically found in commercial products. Consequently, small and medium-sized enterprises must carefully evaluate their technical capabilities before embracing

these cost-effective solutions, balancing the advantages of financial savings with the expertise necessary for effective management and maintenance.

Additional Safe Information Security Measures for SMEs

Given cyberattacks' increasing frequency and sophistication, small and medium-sized organizations (SMEs) must prioritize cybersecurity measures (Chidukwani et al., 2022). Although specific security measures may incur significant expenses, SMEs can choose economical solutions offering substantial protection (Manzoor et al., 2024). By implementing fundamental economic security measures, SMEs may markedly diminish the threat of cyberattacks and protect essential company information (Papathanasiou et al., 2024). Key security steps SMEs can adopt to bolster their cybersecurity posture and preserve critical assets without incurring significant expenses are outlined below.

Passwords

The National Institute of Standards and Technology (NIST, 2020) emphasizes that effective password management is a crucial security measure for small and medium enterprises (SMEs). SMEs should adopt strong, complex passwords to enhance security and prevent unauthorized access. Each password should be unique for individual systems and updated regularly to reduce the risk of compromise. NIST further recommends that passwords include uppercase and lowercase letters, numbers, and symbols while avoiding easily guessable sequences such as "password" or "123456." Moreover, SMEs should implement strict password policies that mandate employees to create robust passwords and prohibit using default or reused credentials. These practices will significantly bolster the overall cybersecurity posture of SMEs.

Mobile Devices

Papathanasiou et al. (2024) highlighted the growing importance of securing mobile devices, as smartphones and tablets have become indispensable tools for small and medium-sized enterprises (SMEs). They contended that, without appropriate security measures, these devices could be entry points for cybercriminals to infiltrate organizational networks. The authors recommended adopting essential security practices, including encryption, secure access controls, and antivirus software, to mitigate this risk. Encryption protects the data on mobile devices, rendering it unreadable to unauthorized users, while antivirus software safeguards against malicious applications and malware.

Furthermore, Borky and Bradley (2018) underscored the necessity for businesses to implement clear policies that mandate employees to secure their devices using strong passwords or biometric authentication. They also advocated for procedures requiring employees to promptly report lost or stolen devices. These measures, they argued, are crucial for decreasing the risk of data breaches and safeguarding organizational assets from cyber threats.

Antivirus Software

Papathanasiou et al. (2024) emphasized the urgent need for Small and Medium Enterprises (SMEs) to adopt up-to-date antimalware software to guard against increasing cyber threats. Antimalware solutions are essential for identifying, mitigating, and eliminating harmful software designed to compromise systems and networks. It relies on a comprehensive database of known malware signatures. The authors noted that routine scans and file access monitoring enable antimalware solutions to identify potential risks and generate alerts for suspicious

activities. Furthermore, Fortinet (2025) and Papathanasiou et al. (2024) underscored the vital prominence of advanced algorithms in identifying new malware by analyzing behavioral patterns and code structures. They contend that without current antivirus software, small and medium-sized enterprises (SMEs) place themselves at significant risk, including exposure to viruses and ransomware that can lead to considerable data loss and other serious repercussions. To mitigate these threats and safeguard sensitive information, SMES must ensure antivirus software is installed on all devices, including computers, servers, and mobile devices.

Firewalls

Bramhe (2024) underscored the essential function of firewalls in protecting an organization’s internal network. Bramhe (2024) underscored the crucial function of firewalls in safeguarding a company's internal network from potential cyberattacks and unauthorized access. Firewalls are crucial safeguards that separate trusted internal networks from untrusted external networks, like the Internet, providing required security. By consistently scrutinizing incoming and outgoing communication by established security protocols, firewalls proved highly successful in identifying and obstructing malicious traffic. Bramhe (2024) illustrated the case of a healthcare SME that effectively deployed a firewall to safeguard its network following an employee's inadvertent engagement with a phishing email. The firewall promptly identified and obstructed the malicious software's connection attempt, averting a possible data breach. This example demonstrates the importance of firewalls as a critical defense mechanism and emphasizes how essential they are to any organization's cybersecurity plan.

Encryption Software

Papathanasiou et al. (2024) and Bramhe (2024) accentuated the essential role of encryption in safeguarding sensitive data within small and medium-sized enterprises (SMEs). According to Papathanasiou et al. (2024), SMEs must establish comprehensive cryptographic policies incorporating robust key management systems and ensure consistent, secure encryption practices across diverse data environments. These policies should encompass several critical components: first, they must align with both organizational objectives and regulatory compliance standards, addressing the protection of data across a range of platforms, including mobile devices, storage media, and data transmissions. Furthermore, such policies should explicitly outline the roles and duties of staff involved in encryption management, guaranteeing responsibility and transparency in enforcement. Equally crucial is carefully selecting and applying appropriate cryptographic standards and algorithms tailored to the SME's specific needs. Lastly, these policies should facilitate the continuous evaluation of encryption's interaction with other security measures, assessing its role within broader organizational security frameworks and ensuring that encryption practices remain adaptable to emerging threats and technological advancements.

Bramhe (2024) also reinforced the necessity of encryption as an essential mechanism for safeguarding sensitive information, including emails, banking records, and consumer information. Encryption prevents unauthorized access by converting data into an unreadable form, especially in cases of data interception or theft. He underscored the need for encryption in transit and at rest to defend against data breaches, physical theft, and system hacking. Both

authors urged SMEs to implement robust encryption systems and efficient key management practices to lower the risks linked with unapproved access to sensitive information.

Multifactor Authentication (MFA)

Bramhe (2024) emphasized that Multifactor Authentication (MFA) is a powerful security measure that requires multiple forms of authentication before granting access to sensitive data or systems. MFA typically includes three elements: something the user knows (e.g., a password), something the user has (e.g., a smartphone), and something the user is (e.g., a fingerprint). This stratified method substantially improved security by complicating illegal access. Bramhe (2024) emphasized the significance of MFA and the essential requirement for securely storing backups in a location distinct from the primary data system to safeguard against potential data loss during cyber catastrophes. Bramhe (2024) demonstrated MFA's efficacy with a compelling case study involving a tiny e-commerce enterprise that retained consumer payment data. Following the implementation of MFA for all administrative accounts, the organization effectively prevented a significant data compromise. A hacker acquired a manager's password via a phishing assault; nevertheless, the second authentication factor thwarted system access, underscoring the vital need for multifactor authentication in protecting against illegal access.

Data Backup

The European Union Agency for Cybersecurity (ENISA, 2020) underscores the importance of regular data backups to ensure business continuity during system failures or cyberattacks, such as ransomware incidents. ENISA advocates for SMEs to implement automated backup systems that consistently restore critical data to external servers or cloud-

based storage, thereby reducing the risk of data loss. Furthermore, the agency recommends storing backups in secure, geographically separate locations from the primary data system to mitigate the impact of cyberattacks further. Following these guidelines helps SMEs protect their operations from ransomware risks and guarantee a quick recovery from system failures, reducing company disturbance and improving general security resilience.

Best Practices for Remote Work

Papathanasiou et al. (2024) emphasized crucial practices for securing remote work, advocating for tailored protocols that effectively address the threats associated with personal data protection. They stress that small and medium-sized enterprises (SMEs) need to set up clear, actionable procedures and ensure employees are well-informed, thoroughly trained, and supported in implementing these protocols. To strengthen security, they recommend tightly controlling remote access to information technology (IT) systems, restricting access to essential tasks, and safeguarding communication through virtual private network (VPN) connections. Key measures include securing wireless fidelity (Wi-Fi) networks with strong passwords and Wi-Fi Protected Access 2 (WPA2) encryption, avoiding storing sensitive data on online platforms, and keeping antivirus software, firewalls, and operating systems current.

Furthermore, Papathanasiou et al. (2024) highlight the need to separate work-related files from personal data, apply encryption to sensitive information, and use secure platforms for teleconferencing. Ultimately, they underscore the critical importance of encrypting files, especially those containing high-risk data, on devices used for telecommuting, as this is essential for maintaining confidentiality and ensuring a robust, secure remote work environment.

Role of Government and Industry Regulations

Rawindaran et al. (2023) contended that governments have a crucial responsibility to enhance cybersecurity for small and medium-sized enterprises (SMEs) through establishing legislative frameworks and enforcing compliance standards. These guidelines help SMEs, which frequently lack the essential resources and know-how, to protect their sensitive data and reduce cyber risks. By enforcing these regulations, governments can create a safer digital environment, promote resilience among SMEs, and strengthen the overall economy.

Similarly, Graham (2025) emphasized that cybersecurity compliance requires adherence to regulatory and industry standards such as the General Data Protection Regulation (GDPR), the Health Insurance Portability and Accountability Act (HIPAA), and the California Consumer Privacy Act (CCPA). These standards guide companies in developing risk-based policies to safeguard sensitive data. Graham (2025) contended that these regulations provide small and medium-sized enterprises (SMEs) with a systematic framework for cybersecurity, mandating the implementation of essential measures like access control, data protection, and incident response preparation.

Compliance safeguards small and medium-sized enterprises from cyber threats and mitigates the financial and reputational risks linked with data breaches and regulatory violations. Since small and medium-sized enterprises typically operate with limited resources and expertise that larger organizations possess, government-mandated rules offer a beneficial framework, enabling them to maintain a robust security posture without developing independent security systems.

Moreover, governments contribute to cybersecurity resilience by promoting cooperation between the public and private sectors (Ikitemur Gokhan et al., 2020). Srinivas et al. (2019) emphasized the significance of cybersecurity regulations in compelling organizations to adopt defensive measures against cyber threats, ensuring a baseline level of security across industries. Mujawar et al. (2025) further emphasized that regulatory frameworks establish security standards and foster transparency and accountability. Public-private partnerships allow SMEs to leverage shared expertise and threat intelligence, helping them stay informed about emerging cyber risks (Lostri et al., 2022). In addition, government-led initiatives, such as cybersecurity awareness programs and funding opportunities, provide SMEs with the support needed to strengthen their security infrastructure (Luna, 2025). By enforcing compliance and encouraging collaboration, governments enable SMEs to build resilience in an ever-evolving digital landscape, reducing their vulnerability to cyberattacks while guaranteeing enduring sustainability in a competitive market.

Technological Trends and Innovations in Cybersecurity for SMEs

Benjamin et al. (2024) emphasize that technological innovations are no longer optional for small and medium-sized enterprises (SMEs); they are essential to circumnavigate the increasing complexity of cyber threats. In an ever-evolving digital landscape, these advancements are vital tools for SMEs to stay ahead of emerging risks. Among the most transformative innovations is artificial intelligence (AI), which, according to Papathanasiou et al. (2024), is critical to bolstering cybersecurity. AI technologies—machine learning, data mining, deep learning, and expert systems—are deemed indispensable in fortifying security frameworks across organizations of all sizes, focusing on SMEs. Utilizing AI skills, SMEs can markedly

improve their capacity to detect and address cyber threats swiftly and accurately, essential in an era of increasingly sophisticated and widespread cyberattacks. The increasing dependence on AI emphasizes the critical necessity for SMEs to embrace advanced technology to protect their sensitive data and ensure operational resilience against rising cyber threats.

Justification for a Qualitative Research Design

Small and medium-sized Enterprises (SMEs) encounter various obstacles when adopting effective cybersecurity practices. The literature review has identified the main barriers, including a lack of education and awareness regarding cybersecurity, financial constraints, and insufficient IT expertise. These challenges and common misconceptions about cyber threats create vulnerabilities that cannot be fully understood through quantitative research alone. Therefore, a qualitative research design is essential to capture the deeper contextual and behavioral factors influencing SMEs' cybersecurity readiness. This strategy enables a comprehensive analysis of the views, motivations, and impediments influencing the uptake of cybersecurity measures.

A fundamental justification for qualitative research is its ability to examine knowledge gaps and awareness deficits in SME cybersecurity practices. Many SME owners and employees lack critical knowledge about cybersecurity fundamentals, such as secure password management, encryption, and phishing prevention. Moreover, gaps in formal training programs often result in employees unknowingly engaging in risky behaviors. While quantitative research can measure the extent of this knowledge gap, qualitative methods, such as in-depth interviews and focus groups, enable researchers to uncover the underlying causes of cybersecurity ignorance, how

awareness is shaped within SMEs, and what specific misconceptions prevent adopting best practices.

Beyond deficiencies in knowledge, small and medium-sized enterprises (SMEs) also encounter financial and resource constraints that hinder their ability to deploy comprehensive cybersecurity strategies. Unlike large corporations with dedicated cybersecurity teams, SMEs frequently lack the staff and financial means to allocate to security infrastructure. Quantitative studies may indicate the percentage of SMEs that allocate minimal budgets to cybersecurity, but they do not reveal the decision-making processes behind these financial choices. A qualitative approach permits for a thorough examination of how SME owners and managers perceive cybersecurity costs, prioritize security investments, and consider alternative solutions. Understanding these perspectives is essential for policymakers and industry stakeholders to develop cost-effective, accessible security measures tailored to SMEs.

In addition to financial barriers, misconceptions about cyber threats significantly influence SME cybersecurity behavior. Many SME owners assume that cybercriminals primarily target large corporations, leading to a false sense of security and minimal investment in cybersecurity defenses. While quantitative research may measure how widespread this belief is, qualitative methods provide the opportunity to explore how and why these misconceptions develop. Interviews and case studies can elucidate how external variables, including industry norms, regulatory pressures, and previous security experiences, can influence small and medium-sized enterprises (SME) decision-making concerning cybersecurity policies and investments.

The real-world impact of cyberattacks on SMEs further highlights the necessity of qualitative inquiry. Cyber incidents can result in financial detriment, operational interruptions, and reputational damage, with their specific effects varying significantly according to the business type and its level of readiness. Quantitative data may quantify average losses, but it does not capture the nuanced effects of cyberattacks on SME business activities, client trustworthiness, and sustained organizational viability. Through qualitative research, firsthand accounts of SME owners and employees can provide rich, contextual insights into how businesses respond to cyber incidents and what lessons they learn in the aftermath.

Another critical advantage of qualitative research is its ability to assess the feasibility and adoption of cybersecurity solutions. Although the literature recommends best practices such as implementing cybersecurity frameworks, investing in affordable security tools, and participating in awareness campaigns, the extent to which these solutions are practical for SMEs remains uncertain. Quantitative research can track adoption rates but does not explain why specific measures succeed or fail within SMEs. By engaging directly with SME stakeholders, qualitative research can uncover barriers to implementation and identify tailored strategies that align with SMEs' specific needs and operational realities.

A qualitative research design is essential for capturing the intricate interplay of education, financial constraints, perceptions, and cybersecurity behaviors in SMEs. Unlike quantitative approaches focusing on numerical trends, qualitative methods offer a deeper, more nuanced exploration of the decision-making processes and contextual factors shaping cybersecurity practices. Researchers can generate actionable insights that inform cybersecurity policies,

training programs, and intervention strategies explicitly designed for SMEs by leveraging interviews and case studies.

Research Framework

Theoretical Framework

The surge in cyberattacks targeting small and medium-sized enterprises (SMEs) can be ascribed to two interrelated factors: a pervasive deficiency of cybersecurity awareness and resource constraints. At the organizational level, small and medium-sized enterprises (SMEs) often fail to recognize their vulnerability as potential targets for cybercriminal activities, resulting in complacency and a reluctance to allocate resources toward cybersecurity investments (Arroyabe et al., 2024; Alahmari & Duncan, 2020). On an individual level, employees within SMEs often demonstrate critically low awareness of cyber threats, further exacerbating vulnerabilities (Zwilling et al., 2022; Taherdoost, 2024). Compounding this challenge, Tetteh (2024) highlights that many SMEs operate with constrained monetary and personnel resources, severely limiting their capacity to implement all-encompassing cybersecurity strategies, ultimately exposing them to a host of cyber threats, including data breaches, operational disruptions, and the risk of complete business failure, underscoring the pressing necessity for targeted interventions and strong policy support.

This study employs a theoretical framework that integrates the Technology Acceptance Model (TAM) with the Technology-Organization-Environment (TOE) framework to address two main challenges: lack of awareness and resource limitations. Utilizing this integrated approach, the study aims to clarify important cybersecurity phenomena, outline critical interrelationships,

and develop predictive insights to guide small and medium-sized enterprises (SMEs) in implementing effective strategies to reduce cyber risks.

The Technology Acceptance Model (TAM)

Developed by Davis (1985), the Technology Acceptance Model (TAM) remains one of the most influential theoretical frameworks for analyzing user acceptance and adoption of emerging electronic technologies and services. At its core, TAM posits that individuals' behavioral intentions (BI) to adopt a given technology are primarily shaped by two critical factors: perceived usefulness (PU)—the degree to which a user believes the technology will enhance their performance—and perceived ease of use (PEOU)—the extent to which they perceive the system as effortless to operate (Bryan & Zuva, 2021). These determinants collectively influence user attitudes, ultimately dictating technology adoption patterns.

Bryan and Zuva (2021) assert that the Technology Acceptance Model (TAM) is essential for identifying the fundamental factors that drive technology adoption. This critical insight not only aids in predicting user behavior but also informs the development of effective strategies for successfully implementing technological innovations. As highlighted by Malatji et al. (2020), TAM underscores the importance of behavioral intentions (BI), which are influenced by two primary determinants: perceived usefulness (PU) and perceived ease of use (PEOU). This core framework thoroughly explains how users interact with emerging technologies. Beyond its theoretical contributions, TAM offers practical value by equipping practitioners with actionable insights for designing pre-implementation strategies that enhance user acceptance and optimize system performance (Marikyan & Papagiannidis, 2024).

The Technology Acceptance Model (TAM) extends beyond traditional technology adoption and has important implications for cybersecurity training. According to Fallatah et al. (2024), establishing a strong security culture is essential for an organization's cybersecurity defense strategy, as it ensures that employees are fully prepared to identify potential threats and respond effectively to mitigate risks. Implementing thorough cybersecurity training programs is a crucial component of this strategy. These programs provide staff with the knowledge and expertise to tackle prevalent cyber threats, including phishing, ransomware, and social engineering attacks (Abrahams et al., 2024; Bada & Nurse, 2019).

A significant obstacle in cybersecurity efforts is the common reluctance among employees to engage in training programs, which significantly diminishes their overall effectiveness (Prümmer et al., 2024). Understanding the factors that affect employee acceptance of such training is vital to overcoming this challenge. Fallatah et al. (2024) point to the Technology Acceptance Model (TAM) as a well-established and evidence-backed framework identifying key factors influencing employees' readiness to adopt cybersecurity best practices. They also recommend incorporating TAM into training strategies as a critical approach to improving employee participation and bolstering organizational security.

The Technology-Organization-Environment (TOE) Framework

The Technology-Organization-Environment (TOE) framework, initially introduced by Tornatzky and Fleischer (1990), remains a foundational model in information systems research. It provides a thorough framework for examining the elements that impact technology adoption, offering valuable insights into how organizations integrate technological innovations (Bryan &

Zuva, 2021). Salmizi et al. (2024) highlight the significance of examining their framework's intricate connections among organizational, environmental, and technological elements. These three interconnected components are essential for organizations to absorb and integrate technological innovations effectively.

Bryan and Zuva (2021) accentuated the three key contexts influenced by the Technology-Organization-Environment (T-O-E) paradigm concerning the adoption and implementation of technological innovations, as follows:

The technological context includes factors influencing adoption at individual, organizational, and industry levels. These factors encompass innovation attributes, system integration, ease of use, trialability, complexity, and direct benefits.

The organizational context encompasses various internal factors, including financial resources, corporate structure, organizational slack, innovation and knowledge capacities, operational competencies, and strategic technological alignment. Additionally, it involves the level of trust within the organization, technological assets, leadership support, encouragement of innovation, the quality of human capital, accumulation of organizational knowledge, expertise, infrastructure, and the organization's overall readiness.

The environmental context includes external factors that influence the business landscape, such as the dynamics of operations, government incentives for growth and sustainability, and the evolving life cycles of industries. These elements are essential for how organizations adapt and succeed in a changing environment.

Extensively employed in empirical research on technology adoption, the Technology-Organization-Environment (TOE) framework remains an essential analytical paradigm, providing scholars and practitioners with a systematic lens to examine the multifaceted interactions among technological innovation, organizational dynamics, and external environmental forces (Daood et al., 2024; Bryan & Zuva, 2021). Its sustained significance highlights its critical role in deciphering the complexities of digital transformation across diverse industries.

An Integrated Framework Combining TOE and TAM Models

The TOE-TAM framework integrates the Technology-Organization-Environment (TOE) framework with the Technology Acceptance Model (TAM) to address the educational deficiencies regarding data security practices within small and medium-sized enterprises (SMEs). This knowledge gap heightens SMEs' susceptibility to cyberattacks, potentially compromising sensitive data and business operations. By analyzing technological, organizational, and environmental elements, the framework provides a means for SMEs to manage cybersecurity risks better and enhance their resilience to evolving cyber threats. It adopts a holistic approach to cybersecurity education, considering the diverse elements that shape SMEs' capacity to effectively mitigate these risks (Bryan & Zuva, 2021; Salmizi et al., 2024).

The framework diagram below offers a refined and detailed visualization of the TAM-TOE models, effectively illustrating each unique component and its intricate relationships within the research context. This representation is both a valuable guide and a significant analytical tool, deepening our comprehension of the operational dynamics and interrelations of these models

inside the study's framework. Each element is thoughtfully designed to illuminate the essential principles and dynamics, making complex theoretical concepts more accessible and intellectually engaging, as outlined below:

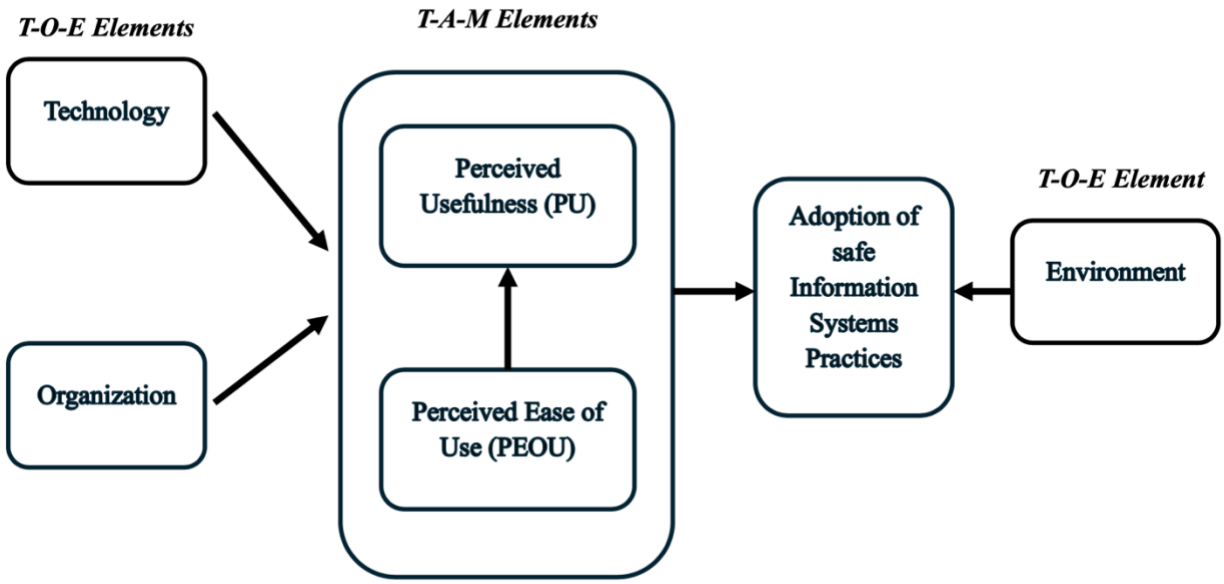

Figure 1. Integrated TAM-TOE Framework

Source: Gangwar, H., Date, H., & Ramaswamy, R. (2015). Understanding determinants of cloud computing adoption using an integrated TAM-TOE model. *J. Enterp. Inf. Manag., 28*, 107-130.

Technological Component – Tailored Cybersecurity Tools and Resources

The technological dimension of the TOE framework is vital for improving cybersecurity education in small and medium-sized enterprises (SMEs) by assessing the accessibility, effectiveness, and adaptability of training tools. Bada and Nurse (2019) highlight that the success

of these programs relies on aligning tools with SMEs' specific needs, ensuring they are user-friendly and convey essential security knowledge. Similarly, Taherdoost (2024) stresses the importance of tailoring training platforms to fit SMEs' unique challenges, such as varying learning preferences, knowledge levels, limited IT proficiency, and financial constraints. By identifying gaps in current training resources, the TOE framework can enhance training technologies, boost employee awareness, and reduce cybersecurity vulnerabilities (Hasani et al., 2023). Ultimately, the design and usability of training platforms are crucial for strengthening the cybersecurity resilience of SMEs.

Organizational Component – Internal Organizational Support and Readiness

The organizational aspect of the TOE-TAM framework investigates the internal frameworks, resources, and culture within small and medium-sized enterprises (SMEs), focusing on challenges such as limited financial resources and insufficient leadership commitment to cybersecurity education (Bryan & Zuva, 2021). It assesses organizational preparedness, employee involvement, and the distribution of resources for cybersecurity training, emphasizing obstacles like inadequate leadership backing or financial limitations that may undermine the effectiveness of awareness initiatives (Wallace et al., 2021). Understanding these internal dynamics is crucial for bridging knowledge gaps and ensuring active leadership involvement in supporting training endeavors.

Environmental Component – External influences and industry context

The environmental factor in the TOE-TAM framework emphasizes the significance of external factors affecting cybersecurity practices in small and medium-sized businesses (SMEs).

Key elements include market forces, industry standards, and legal requirements, all of which significantly impact the development and delivery of cybersecurity training. Such training is essential for compliance with legal and industry standards (Bryan & Zuva, 2021; Wallace et al., 2021). By incorporating an external viewpoint, the framework evaluates how effectively small and medium-sized enterprises (SMEs) respond to emerging cyber threats and regulatory shifts. This assessment ensures that their cybersecurity education programs align with current industry standards. By ensuring this alignment, SMEs bolster their protection against current cybersecurity threats and meet broader industry standards, hence reducing the probability of legal complications and reputational damage.

Technology Acceptance Model (TAM) – User Perception and Adoption

The TAM component of the framework concentrates on comprehending employee perspectives of cybersecurity training initiatives, particularly their perceived usefulness and ease of use. The effectiveness of these programs is deeply influenced by how employees view them, with positive perceptions fostering higher engagement and better retention of security knowledge. By incorporating TAM, the TOE-TAM framework enables SMEs to assess how training design and delivery impact employee participation and engagement. This understanding allows organizations to refine training content and delivery methods to improve user adoption, ultimately strengthening the effectiveness of their cybersecurity initiatives. This helps reduce vulnerabilities associated with inadequate cybersecurity knowledge among employees (Bryan & Zuva, 2021).

Comprehensive Assessment Through the Integration of TOE and TAM

The TOE-TAM model synergizes the strengths of the Technology-Organization-Environment (TOE) framework and the Technology Acceptance Model (TAM) to offer small and medium-sized enterprises (SMEs) a holistic approach for assessing their cybersecurity training initiatives. This unified framework includes key elements such as technological appropriateness, organizational backing, environmental influences, and user adoption. By considering these elements collectively, SMEs can pinpoint weaknesses in their training initiatives and assess whether these programs align with their organizational objectives and external pressures. This multifaceted evaluation empowers SMEs to enhance their cybersecurity programs, effectively addressing knowledge gaps and vulnerabilities, thereby reducing their susceptibility to cyberattacks (Bryan & Zuva, 2021).

Continuous Feedback and Iteration

A key advantage of the TOE-TAM framework is its ability to support continuous feedback and iterative improvement. As cybersecurity threats constantly evolve, this adaptable approach allows small and medium-sized enterprises (SMEs) to adjust their training programs over time based on employee input and changing threat environments. This iterative process helps SMEs enhance and update their cybersecurity education, ensuring the training remains relevant and practical. By incorporating regular feedback loops, the framework enables organizations to stay proactive against emerging threats and adjust their strategies to maintain resilience against evolving cyber risks (Bryan & Zuva, 2021).

Customization and Scalability for SMEs

The TOE-TAM framework benefits SMEs, as they frequently encounter resource limitations and difficulties in expanding cybersecurity efforts. By allowing customization based on an organization's specific needs and capacities, the framework ensures that SMEs can tailor their cybersecurity education programs to their unique resources, technological infrastructure, and operational constraints. This flexibility enables even resource-limited SMEs to implement practical cybersecurity training, maximizing impact while addressing limitations such as small budgets or limited staff. The framework's scalability ensures that SMEs can develop sustainable, adaptable training programs that meet immediate needs and long-term cybersecurity goals (Bryan & Zuva, 2021).

The TOE-TAM framework serves as an essential guide for small and medium-sized enterprises (SMEs) seeking to enhance their cybersecurity training initiatives. SMEs can pinpoint and address deficiencies in their cybersecurity knowledge by analyzing the intricate relationships among technological capabilities, organizational readiness, environmental factors, and employee perceptions. This flexible and responsive framework tackles existing challenges and anticipates future needs, empowering SMEs with the agility and confidence to navigate the constantly changing digital landscape. By utilizing the TOE-TAM framework, SMEs can cultivate a skilled workforce, minimize risk exposure, and foster a culture of ongoing enhancement. Ultimately, this creates a solid foundation for a sustainable cybersecurity strategy that strengthens resilience and protects organizational integrity against ongoing cyber threats.

Summary of Literature Support

The escalating prevalence of cyberattacks targeting small and medium-sized enterprises (SMEs) underscores an urgent imperative for bolstering cybersecurity awareness and education (Chidukwani et al., 2022; Arroyabe et al., 2024; Bada & Nurse, 2019). As key drivers of economic growth and innovation (Algan, 2019), SMEs frequently function under significant financial and technological constraints, making them especially vulnerable to cyber threats. Their limited cybersecurity expertise exacerbates these risks, leaving them open to exploitation by increasingly sophisticated cybercriminals (Tetteh, 2024; Arroyabe et al., 2024). The ramifications of such attacks transcend financial losses, resulting in significant operational interruptions and eroding stakeholder trust. Furthermore, they threaten the security of sensitive corporate and consumer data, compounding the long-term impact of inadequate cybersecurity measures (Clearly, 2023; Woodgate, 2024; Kindle, 2025). A significant contributor to this increased vulnerability is the prevalent absence of cybersecurity education and awareness among SME owners and staff, which reinforces structural deficiencies and amplifies susceptibility to emerging digital threats (Prümmer et al., 2024; Chaudhary et al., 2023; Taherdoost, 2024).To address this, targeted training programs, evidence-based policy interventions, and robust security frameworks must be implemented to strengthen SMEs' resilience against the growing cyber threat landscape.

Despite the growing recognition of cybersecurity risks, numerous small and medium-sized enterprises (SMEs) remain largely unprepared due to insufficient training efforts and the absence of thorough awareness programs (Chidukwani et al., 2022; Arroyabe et al., 2024; Bada & Nurse, 2019). Mitigating these vulnerabilities necessitates a deep comprehension of SMEs'

unique cybersecurity challenges and theoretical frameworks supporting adequate security practices. This literature review offers a critical examination of existing research on SME cybersecurity, focusing specifically on the vital roles of training, awareness, and education in enhancing digital resilience.

This review establishes a strong basis for identifying and evaluating strategic interventions aimed at enhancing SMEs' cybersecurity capabilities through the integration of pivotal research, theoretical insights, and methodological frameworks. Ultimately, emphasize the urgent necessity for specialized educational programs and data-informed training initiatives that provide SMEs with the knowledge and tools required to safeguard their digital infrastructures against an evolving and hostile cyber threat landscape.

Cybersecurity Challenges Faced by SMEs

Zwilling et al. (2022), Nurse (2021), Abrahams et al. (2024), and Chidukwani et al. (2022) present substantial evidence of the increased susceptibility of small and medium-sized enterprises (SMEs) to cyber threats, attributing this susceptibility primarily to insufficient cybersecurity awareness and constrained financial and human resources. Their research underscores SMEs' systemic challenges in adopting robust security measures, emphasizing that inadequate investment in cybersecurity infrastructure leaves them disproportionately exposed to increasingly sophisticated cyberattacks. By highlighting the critical gaps in SME cybersecurity preparedness, these scholars establish the urgent need for comprehensive, scalable interventions to mitigate risk and enhance digital resilience.

Arroyabe et al. (2024) and Alahmari and Duncan (2020) further illuminate a fundamental issue in SME cybersecurity: the prevalence of a false sense of security. Their findings indicate that numerous SMEs substantially underestimate their vulnerability to cyber hazards, resulting in a reactive rather than a proactive cybersecurity strategy. This misperception leaves businesses ill-equipped to defend against data compromises, monetary losses, and interruptions to their operations. Without a paradigm shift in how SMEs perceive and address cybersecurity threats, they will remain prime targets for cybercriminal activity. The work of these researchers underscores the pressing need for targeted awareness initiatives and proactive risk mitigation strategies to fortify SMEs against the evolving cyber threat landscape.

Zwilling et al. (2022), Taherdoost (2024), Arroyabe et al. (2024), and Papathanasiou et al. (2024) underscore a critical weakness in SME cybersecurity: the pervasive lack of formal training among employees, which significantly exacerbates organizational vulnerabilities. Their research reveals that many SME employees struggle to spot and react effectively to common cyber threats, including phishing attacks, malware intrusions, and social engineering tactics. This deficiency in cybersecurity literacy not only increases the likelihood of inadvertent security breaches but also weakens the overall security posture of SMEs. Without structured education programs integrating theoretical foundations and practical threat response mechanisms, SMEs remain highly susceptible to cyber exploitation. These findings reinforce the pressing demand for targeted educational programs designed to foster a cybersecurity-conscious environment, thereby enhancing SMEs' perseverance against an increasingly sophisticated threat landscape.

Financial and human resource constraints and deficiencies in cybersecurity training pose significant challenges to implementing cybersecurity measures in small and medium-sized

enterprises (SMEs). Tetteh (2024) highlights that numerous SMEs, constrained by tight budgets, find it challenging to dedicate funds for strong safety measures or hire dedicated cybersecurity staff. Consequently, these businesses often rely on outdated and inadequate protective measures, substantially increasing their exposure to cyber risks. This persistent underinvestment intensifies their vulnerability and leaves SMEs unprepared to confront the increasingly complex and evolving cyber threats they face. Tetteh's study accentuates the critical necessity for affordable, adaptable security structures tailored to the specific operational limitations of SMEs. Addressing these systemic gaps necessitates a comprehensive methodology that includes targeted cybersecurity education, policy interventions, and innovative technological innovations to enhance a more resilient and sustainable cybersecurity posture within the SME sector.

Theoretical Framework: TAM and TOE Integration

This study adopts a dual-theoretical approach to tackle the cybersecurity issues faced by small and medium-sized enterprises (SMEs), integrating the Technology Acceptance Model (TAM) and the Technology-Organization-Environment (TOE) framework. The TAM, developed by Davis (1989), provides a psychological and behavioral lens on technology adoption, focusing on perceived usefulness and ease of use as critical aspects that impact decision-making processes. This model highlights how individual perceptions shape the willingness of SMEs to embrace new cybersecurity technologies.

A growing collection of studies highlights the decisive role of usability in adopting cybersecurity measures within SMEs. Studies by Bhad (2025), Benjamin et al. (2024), Tetteh (2024), and Bada et al. (2019) reveal that SMEs are significantly more likely to implement

security technologies when these solutions are intuitive, minimally disruptive, and critical for ensuring business continuity. This finding underscores the pressing need for cybersecurity frameworks that are effective and seamlessly integrated into SME operations. Such integration is essential for fostering higher adoption rates, strengthening cyber resilience, and securing the long-term viability of security approaches. Prioritizing usability and alignment with operational needs enables cybersecurity solutions to bridge existing gaps and fortify SMEs' defenses against evolving cyber threats.

The TOE framework, introduced by Tornatzky and Fleischer (1990), complements this analysis by considering internal and external factors influencing technology adoption. Research by Bhad (2025), Auyporn et al. (2023), and El-Hajj and Mirza (2024) emphasizes the crucial role of regulatory compliance, competitive pressures, and organizational readiness in shaping cybersecurity investment decisions within SMEs. This approach elucidates the intricate interaction of technological capabilities, organizational culture, and external market dynamics, offering a deeper insight into the factors influencing cybersecurity practices.

Through the fusion of the TAM and TOE frameworks, this study provides a multidimensional perspective on cybersecurity adoption within SMEs. The amalgamation of these frameworks allows for a holistic comprehension of how personal views on cybersecurity tools interact with wider organizational and environmental influences. Ultimately, this theoretical synergy enhances the ability to predict cybersecurity adoption behaviors, offering valuable insights for developing targeted, context-specific strategies to promote cybersecurity adoption in resource-constrained SME environments.

Research Methodology and Justification

Qualitative research methodologies have become indispensable in cybersecurity research, offering invaluable comprehension of the underlying processes of organizational behaviors and security practices. Goldberg and Allen (2015) emphasize the pivotal role of qualitative case studies in uncovering rich, context-specific details about cybersecurity challenges, enabling a deeper understanding of the complex interplay among technological, human, and organizational factors. In alignment with this approach, the present study conducts a comprehensive case analysis of a small to medium-sized enterprise's (SME) cybersecurity posture, aiming to identify critical vulnerabilities, assess risk perceptions, and explore effective mitigation strategies.

This study utilizes a purposive sampling method to facilitate a comprehensive analysis, specifically focusing on essential stakeholders within the organization's cybersecurity framework. This include IT personnel, executives, and frontline employees, whose direct participation in cybersecurity decision-making and execution guarantees that their perspectives are precisely recorded and represented. This targeted selection method facilitates a comprehensive understanding of cybersecurity strategies from various corporate viewpoints.

In addition to purposive sampling, a mixed-methods data collection strategy is utilized, increasing the study's rigor. By combining semi-structured interviews, online surveys, and document analysis, the research adheres to the best practices for cybersecurity research as described by Yin (2018). This multifaceted methodology offers a comprehensive view of cybersecurity dynamics within SMEs, fostering a more profound understanding of the associated challenges and strategic responses. Additionally, triangulation—comparing data from various

sources—enhances the trustworthiness and dependability of the results, ensuring a solid and multifaceted view of the organization's cybersecurity environment.

This research adds to the scholarly discussion on SME cybersecurity through these rigorous and strategically chosen methodologies. It provides practical strategies for enhancing cybersecurity practices, offering actionable insights to drive meaningful improvements in SMEs' security frameworks.

Contribution to the Body of Knowledge

This literature review serves as a comprehensive scholarly exploration, synthesizing existing studies to build a robust intellectual framework for the research. A critical examination of cybersecurity challenges, theoretical paradigms, and methodological approaches offers a structured and thorough perspective on analyzing and addressing the weaknesses faced by small and medium-sized enterprises (SMEs) in their cybersecurity practices.

A substantial contribution of this study is its innovative amalgamation of the Technology Acceptance Model (TAM) and the Technology-Organization-Environment (TOE) framework. This integration provides a novel viewpoint on how SMEs navigate the delicate balance between adopting cybersecurity measures and managing operational constraints. This approach enriches theoretical discourse by emphasizing the dynamic interplay between individual technology acceptance and broader organizational limitations. It provides important perspectives on the decision-making processes that influence cybersecurity practices.

Furthermore, adopting a qualitative case study methodology adds significant empirical depth to the investigation, facilitating a detailed examination of SMEs' real-world cybersecurity challenges. This approach captures the lived experiences of SME stakeholders, generating actionable insights that can directly influence policy development and practical interventions.

The review highlights the pressing need for enhanced cybersecurity awareness within SMEs while presenting a theoretical and methodological roadmap for future research. This review lays the foundation for a transformative investigation of scalable, practical cybersecurity solutions tailored to SMEs' unique constraints and realities by addressing key research gaps and providing a comprehensive analytical framework.

SECTION 3: Research Methodology

Role of the Researcher, Background, and Experience

The role of a researcher in qualitative inquiry is complex, requiring a balance between engagement and objectivity (Creswell & Poth, 2018). Researchers must understand participants' thoughts and experiences while maintaining some distance from the study's phenomenon (Denzin & Lincoln, 2018). This balance necessitates methodological expertise and reflexivity to prevent personal biases from influencing data interpretation (Jamieson et al., 2023). Skilled researchers use techniques like in-depth interviews and field observations while analyzing data within a structured framework (Creswell & Creswell, 2023; Khan et al., 2023). Immersing themselves in participants' experiences helps them construct meaning from the data (Tracy, 2020). However, this deep involvement can lead to bias, so ongoing self-monitoring and a commitment to ethical standards are crucial to maintaining the credibility of the findings (Leach & Welfel, 2018).

For this research project, a robust foundation in research methodology developed during MBA studies is leveraged, including the completion of a thesis that greatly enhanced research design and analysis skills. Proficiency is demonstrated in synthesizing information from a wide range of sources, systematically identifying and extracting key insights to improve the depth and accuracy of analysis (Yin, 2018). A strong commitment to accuracy is reflected in meticulous fact-checking practices, ensuring the integrity and reliability of findings (Silverman, 2020).

Additionally, the researcher's multidisciplinary background facilitates the integration of concepts across various fields, thereby enriching the depth and scope of the research (Lim,

2024). These competencies enable the researcher to deliver valuable insights that support informed decision-making and contribute to the strategic growth of all stakeholders.

Population and Sampling

Casteel and Bridier (2021) stress the importance of clearly defining the target population in any research endeavor, as this ensures the study focuses on the group most pertinent to its research questions. As delineated by the authors, the study population includes various individuals, dyads, groups, organizations, or other entities researchers want to investigate and comprehend. This broad collective serves as the primary focus of the research, allowing for the generalization or application of findings to similar groups or entities. In contrast, the target population is more specifically defined, shaped by specific criteria and accessibility, and represents the subset from which the study sample is drawn. Recognizing the distinction between the broader study population and the more narrowly defined target population is crucial for maintaining the relevance and focus of the research, ultimately yielding meaningful and applicable insights (Gupta et al., 2019).

This research examines how cybersecurity awareness and training play a role in ensuring sustained business success in small and medium-sized enterprises (SMEs). While the broader population of interest encompasses SMEs that may be affected by or benefit from cybersecurity initiatives, the study specifically targets the employees of K2 Partnering Solutions, a Boston-based SME. These individuals are directly involved with or impacted by the company's cybersecurity training and awareness programs. Their involvement is intentional, as they are relevant to the study's objectives and accessible to the researcher, as Willie (2024) noted. By

focusing on this specific group, the study seeks to generate insights that will illuminate the experiences of K2 Partnering Solutions and provide valuable implications for other SMEs facing similar challenges and opportunities in leveraging cybersecurity to enhance their business resilience and performance.

Sampling Frame and Representativeness

Lee and Landers (2022) define the sampling frame as a subset of the target population from which data is gathered. This subset is determined by the study's scope and logistical constraints, and it is chosen based on practical accessibility and its relevance to the research objectives.

This study investigates the role of cybersecurity awareness in enhancing the long-term performance of small and medium-sized enterprises (SMEs). The sampling frame comprises key business leaders and employees at K2 Partnering Solutions who are actively involved in cybersecurity initiatives. While the broader target population includes all employees at K2 Partnering Solutions, the sampling frame specifically highlights those engaged with or affected by the cybersecurity training programs. This focused approach ensures that the data gathered is closely related to the research questions and aligns with the study's main goals.

The significance of the sampling frame stems from its emphasis on participants who possess firsthand experience with the company’s cybersecurity programs. Key business leaders, including the CEO, CFO, and IT directors, along with employees from departments such as IT security, human resources, and operations management, will be selected for their comprehensive understanding of the effectiveness of these programs. These individuals play vital roles in

designing, implementing, and evaluating cybersecurity initiatives, making them well-equipped to provide rich, contextually relevant data. Focusing the sampling frame on these participants, the study seeks meaningful insights into how cybersecurity training influences business performance, particularly within small and medium-sized enterprises (SMEs).

The representativeness of the sampling frame is emphasized by its alignment with the characteristics of small and medium-sized enterprises (SMEs) engaged in similar cybersecurity initiatives. The selected business leaders and key employees reflect the experiences of other SMEs implementing cybersecurity training programs, making the findings pertinent to a broader array of organizations. For example, insights from C-suite executives, such as the CEO and Chief Information Security Officer (CISO), will shed light on strategic decision-making. At the same time, IT security personnel and operational managers will provide valuable perspectives on cybersecurity initiatives' practical implementation and effectiveness. This diversity within the sampling frame ensures that the study's conclusions will be relevant to many SMEs grappling with cybersecurity challenges.

Sampling Method and Rationale

Sampling methods are critical in determining the population from which data are collected, directly influencing research findings' relevance and applicability (Ahmed, 2025). An appropriate sampling strategy is essential to ensure alignment with the research objectives and the target population's characteristics. According to Lee and Landers (2022) and Ahmed (2024), sampling methods are broadly classified into two main categories: probability sampling and non-probability sampling. Probability sampling guarantees that each member of the population has an

equal likelihood of selection, thereby reducing bias and improving the generalizability of the findings. In contrast, non-probability sampling is more selective, often guided by the researcher's judgment or practical convenience. It is particularly suitable for exploratory or qualitative research where the goal is to gain in-depth, context-specific insights rather than generalize findings to a broader population.

This study will utilize purposive or judgmental sampling, wherein participants are intentionally selected based on specific characteristics or expertise pertinent to the research objectives (Bazen et al., 2021). This method is particularly suited to qualitative research, which aims to gather in-depth, nuanced insights from a targeted group rather than striving for a representative sample. Bazen et al. (2021) noted that purposive sampling ensures that participants possess the requisite knowledge or experience, enhancing the findings' relevance and depth.

This study will focus on a purposive sample of employees from K2 Partnering Solutions who possess firsthand experience with the company’s cybersecurity training programs. Due to their direct involvement with cybersecurity initiatives, these individuals are uniquely positioned to provide valuable insights into how cybersecurity awareness impacts business performance. The choice of purposive sampling aligns with the research objective of exploring how cybersecurity awareness contributes to the long-term resilience and success of small and medium-sized enterprises (SMEs). By focusing on individuals most intimately engaged with the company’s cybersecurity efforts, the study guarantees that the data gathered will be contextually relevant and directly applicable to the research questions.

This study will employ purposive sampling to deliberately select employees at K2 Partnering Solutions who possess direct, operational experience with the organization's cybersecurity training programs. These participants, by virtue of their active involvement in cybersecurity initiatives, are particularly well-positioned to offer contextually rich insights into the ways in which cybersecurity awareness affects business performance. This sampling strategy is purposefully aligned with the study's overarching aim: to investigate the role of cybersecurity awareness in promoting the long-term resilience and success of small and medium-sized enterprises (SMEs). To achieve a comprehensive and multidimensional understanding of the phenomenon under investigation, data will be collected through a triangulated methodology comprising semi-structured interviews (Appendix B), an online survey questionnaire (Appendix C), and document analysis (Appendix D). By engaging participants who are most directly connected to the research focus, the study is designed to generate analytically robust, nuanced, and practically relevant findings.

The primary strength of purposive sampling lies in its ability to select individuals most likely to offer meaningful and contextually relevant insights (Campbell et al., 2020). This method enhances the depth and specificity of the data, ensuring that the study effectively captures the nuances of the topic under examination. Additionally, purposive sampling maximizes resource efficiency by enabling the researcher to concentrate on a smaller, more pertinent group of participants, thus eliminating the need for a broader, less focused sample (Ahmed, 2024). This targeted approach ensures that the study's findings are both insightful and tightly aligned with its objectives, ultimately making a significant contribution to the field of cybersecurity awareness in SMEs.

Sample Size Determination

Selecting the right sample size is a crucial aspect of research design, especially in qualitative studies, as it directly influences the reliability, depth, and validity of the results. A critical factor in determining the ideal sample size is saturation, the point at which additional data collection no longer yields new insights or themes (Bazen et al., 2021). Studies suggest that saturation is usually achieved with a relatively small group of participants, with the range typically falling between 9 and 17 detailed interviews, depending on the study's design and goals (Hennink & Kaiser, 2022). This approach ensures that the data gathered is thorough enough to answer the research questions without redundant repetition.

A sample size of 10 respondents has been purposefully selected to achieve saturation, ensuring the data gathered is rich and diverse enough to meet the study goals. This sample size balances obtaining the necessary depth of insights for a comprehensive analysis and addressing practical constraints such as time, resources, and logistics. It is expected to yield a robust dataset that facilitates the acquisition of detailed insights while maintaining the reliability and practicality of the results within the study's scope. By following the saturation principle, this research seeks to establish a reliable basis for drawing valid and pertinent conclusions (Bazen et al., 2021; Hennink & Kaiser, 2022).

Inclusion and Exclusion Criteria

The inclusion and exclusion criteria ensure that the sample appropriately reflects the target population and aligns with the study's objectives. Inclusion criteria specify the attributes

necessary for participant eligibility, while exclusion criteria outline the factors that disqualify specific individuals from participation (Pawlik & Sosa, 2020; Nikolopoulou, 2022).

For this study, the inclusion criteria require that participants be employees directly related to cybersecurity, IT, or departments responsible for managing sensitive data, such as HR, Finance, or Marketing. Participants should have engaged with the company's cybersecurity training or programs to ensure their experiences are applicable to the research. Additionally, participants must both willing and capable of participating in the study.

The exclusion criteria encompass employees whose roles do not handle sensitive data or contribute to cybersecurity initiatives. Those without experience or knowledge of the company's cybersecurity practices will also be excluded, as their participation would not significantly enhance the research. Furthermore, individuals who are disengaged or unwilling to provide informed consent will not be eligible for inclusion in the study.

Demographic Characteristics

Although the main emphasis of this study is on the professional roles and experiences relevant to the research questions, it is essential to recognize the demographic attributes of the chosen sample, such as age, gender, and ethnicity, as they may offer important context for the results. While these factors will not be the central focus of the analysis, ensuring demographic diversity within the sample may offer broader insights into how varied perspectives contribute to the research outcomes. The study will emphasize diversity in professional backgrounds, ensuring adequate representation across key organizational departments, including IT, HR, and Finance. Additionally, participants must have a foundational understanding of data security principles,

ensuring they are well-equipped to provide meaningful and informed contributions to the study's exploration of cybersecurity awareness and its impact on organizational performance.

Challenges and Limitations

Several challenges and limitations may impact the sampling process. A significant challenge is ensuring all participants fulfill the specific experience and role-related criteria essential to the study. Some targeted employees may have limited exposure to cybersecurity practices, which could affect the depth and comprehensiveness of the data gathered. Furthermore, legal and security constraints, such as non-disclosure agreements or confidentiality protocols, may restrict participation, narrowing the pool of eligible respondents. These factors could limit the diversity of perspectives and affect the sample's representativeness, potentially influencing the study's findings.

Ethical Considerations

Ethical considerations are fundamental to maintaining the integrity and credibility of any research endeavor (Correia, 2023; Taquette & Borges Da Matta Souza, 2022). In the present study, informed consent (see Appendix A) will be obtained from all participants, ensuring that they are comprehensively informed about the study's objectives, methodologies, and any potential risks involved. Participants will be explicitly apprised of their right to withdraw from the study at any time without incurring any negative consequences. Stringent measures will be implemented to uphold confidentiality, including secure data storage and the anonymization of all identifying information to protect participants' identities and sensitive data. These ethical

protocols are designed to ensure the respectful treatment of participants and to safeguard their rights throughout the research process.

This section on population and sampling underscores the critical distinction between the broader study population and the more narrowly defined target population, shaped by specific criteria of relevance and accessibility. The research specifically focuses on small and medium-sized enterprises (SMEs), narrowing its scope to employees of K2 Partnering Solutions in Boston who are directly involved in or impacted by cybersecurity training initiatives. This targeted approach ensures the study's relevance, providing deep insights that are particularly beneficial for SMEs navigating similar cybersecurity challenges. The sampling frame comprises key individuals with firsthand experience in cybersecurity programs, directly aligning with the study's objectives and facilitating the collection of rich, contextually relevant data.

The study utilizes a purposive sampling method, selecting individuals based on specific traits or expertise relevant to the research question. This approach ensures that the data collected is comprehensive and closely aligned with the study's objectives. A sample size of 10 respondents has been chosen to achieve data saturation, striking a balance between obtaining in-depth insights and addressing practical constraints.

Inclusion and exclusion criteria are rigorously defined to ensure that the sample accurately represents the target population, focusing on participants in cybersecurity and data security roles. While demographic diversity is considered, the primary emphasis is placed on professional roles and relevant experiences, ensuring that the study's findings are robust and broadly applicable to SMEs across various sectors.

Data Collection and Analysis

Creswell and Poth (2018) highlight that qualitative research is particularly effective for exploring complex social phenomena by capturing the depth and diversity of participants' lived experiences. In alignment with this paradigm, the present study adopts a qualitative case study design to investigate cybersecurity awareness and practices within K2 Partnering Solutions. Semi-structured interviews (see Appendix B), as recommended by Fusch, Fusch, and Ness (2018), Bhandari (2023), and Lim (2024), are employed to elicit rich, contextually grounded insights from key stakeholders. These interviews are complemented by online surveys (Appendix C), which enhance accessibility and encourage candid engagement, and by document analysis (Appendix D), which provides essential contextual depth through the systematic examination of internal policies and procedural records. Together, these triangulated methods form a robust and coherent research strategy that directly supports the study's research questions, enabling a comprehensive and integrated understanding of the organization's cybersecurity culture, behaviors, and vulnerabilities.

Data Collection Instruments

This study utilizes a triangulated approach comprising three complementary data collection methods: semi-structured interviews (Appendix B), an online survey questionnaire (Appendix C), and a document review checklist (Appendix D). This multi-method strategy is designed to enhance methodological rigor and ensure a comprehensive depth of inquiry. Anchored in the methodological paradigms articulated by Silverman (2020) and Creswell and Poth (2018), these instruments collectively enable a thorough examination of organizational

cybersecurity practices. The semi-structured interview protocol, informed by contemporary cybersecurity scholarship, incorporates open-ended questions that solicit rich, experience-based narratives from participants. This approach facilitates the capture of complex and nuanced organizational perspectives, thereby enriching the overall analytical framework (Vacca, 2021; Tisdell, 2025).

The online survey incorporates both open- and closed-ended questions to capture a combination of qualitative and quantitative insights. To enhance clarity and relevance, it will be pilot tested with two staff members from K2 Partnering Solutions, thereby supporting the reliability of the instrument (Fusch, Fusch, & Ness, 2018). In parallel, a structured document review checklist will facilitate the systematic examination of internal policies, training logs, and incident reports, offering critical contextual data. This triangulated approach enhances the depth of the investigation into cybersecurity behaviors and protocols, while also strengthening the study's credibility and validity through the cross-verification of data from multiple sources (Swanagan, 2024; U.S. Department of Health and Human Services, 2022; Yin, 2018).

Participants and Sample

This study will use a purposive sampling strategy to select ten participants from across various departments within K2 Partnering Solutions. Participants will be selected for their ability to provide informed insights into the organization's cybersecurity practices, with a particular focus on internal policies and training programs (Moser & Korstjens, 2017; Vasileiou et al., 2018). The sample will include IT professionals, administrative staff, departmental employees, and department heads who have been with the company for at least six months and are directly

involved in cybersecurity protocols. This approach ensures diverse perspectives across both functional roles and hierarchical levels within the organization.

The sampling frame will target key personnel who manage, implement, or are impacted by cybersecurity training efforts. This includes high-level executives such as the CEO and CFO, alongside IT directors, HR managers, and operations staff. By focusing on participants directly engaged in cybersecurity activities, the study aims to generate a comprehensive understanding of behaviors, attitudes, and vulnerabilities within a small-to-medium enterprise (SME) context. This approach is grounded in established qualitative research methodologies (Hennink & Kaiser, 2021; Rahimi & Khatooni, 2024; Saunders et al., 2017) and aligns with current literature emphasizing the importance of cybersecurity leadership and training effectiveness in SMEs (Khan et al., 2025; Writer, 2024; Arroyabe et al., 2024).

Data Collection Procedures

This study adopts a structured and ethically rigorous data collection framework to ensure methodological validity and the protection of participant rights. After obtaining institutional review board (IRB) approval, prospective participants will be contacted via email and given an information sheet along with an informed consent form. These materials explain the study's objectives, data collection procedures, and ethical safeguards, including confidentiality, voluntary participation, and the right to withdraw without consequence. Only individuals who provide explicit consent will participate, aligning with ethical standards in qualitative research (Creswell & Poth, 2018).

The initial data collection phase includes semi-structured interviews, arranged flexibly and held either in person or through secure, encrypted video conferencing. Each interview will last 10 to 15 minutes and be audio-recorded with the participant's permission. A standardized interview guide ensures consistency while allowing open-ended dialogue that captures rich, context-specific insights into cybersecurity practices and perceptions (Adams & Lawrence, 2018). This qualitative method supports exploring individual experiences and interpretations, which is essential for understanding complex organizational dynamics.

Concurrently, an online questionnaire (Appendix C) will be distributed via a secure, password-protected platform and will remain accessible for two weeks to promote participant engagement. The questionnaire will feature a mix of open-ended and closed-ended questions, allowing for the capture of descriptive responses and measurable patterns while ensuring participant anonymity. In the final phase, internal records—such as cybersecurity policies, training documentation, and incident logs—will be examined using a standardized checklist (Appendix D). The company's compliance officer will facilitate access. Integrating interviews, surveys, and document analysis, this triangulated approach strengthens validity and offers a multidimensional perspective on cybersecurity at K2 Partnering Solutions (Fusch, Fusch, & Ness, 2018).

Data Recording, Storage, and Management

To maintain the highest standards of research ethics and data management, this study will enforce strict protocols for recording, storing, and handling data in full accordance with institutional and regulatory requirements. All interview data will be audio-recorded with

participant consent and transcribed verbatim to preserve the authenticity and richness of their narratives. These transcripts and the original audio files will be stored on encrypted, password-protected digital drives, accessible only to the principal investigator. This access restriction is essential to maintaining the confidentiality of sensitive information and protecting participant identities (Saunders et al., 2019).

Survey submissions will be automatically de-identified upon completion and stored on secure servers managed by trusted third-party providers. A unique identification code will be assigned to each participant, ensuring that their responses remain untraceable to their personal identities. Backup copies of all data files will be securely stored on encrypted external storage systems to reduce the risk of data loss or corruption. Additionally, a thorough audit trail—documenting data handling, access, and modifications—will be carefully maintained to ensure clarity, accountability, and the integrity of the research process (Tracy, 2020). These rigorous data management practices will uphold ethical principles while safeguarding the confidentiality, security, and accuracy of the data throughout the study.

Data Analysis Plan

Given the qualitative design of this study, thematic analysis will be employed to examine the data. This approach is well-suited for identifying recurring patterns and themes within textual data, enabling a rich and nuanced interpretation of participant perspectives (Labra et al., 2019; Caulfield, 2023). Data collected through semi-structured interviews, online surveys, and internal document reviews will be systematically coded and organized to support the development of meaningful themes.

The analysis will begin with open coding to generate initial codes from key phrases, concepts, and repeated patterns across the dataset (Allsop et al., 2022; Limna, 2023). Initial codes may include phrases such as *"only get trained once a year," "not sure who to report to,"* or *"cybersecurity is not part of my job,"* which highlight underlying patterns of disengagement, role ambiguity, and lack of clarity around responsibilities. These codes will then be grouped during the axial coding phase into broader categories that reflect the study's primary themes, including cybersecurity awareness, training effectiveness, and policy enforcement (Siegle, 2023). For example, references to infrequent communication, uncertainty about reporting channels, and outdated training materials may be clustered under the axial code "fragmented cybersecurity culture." Thematic maps will visually illustrate the connections between these categories, facilitating interpretation of the results. Finally, the themes will be analyzed concerning the study's research questions and existing literature to provide a clear narrative of organizational cybersecurity practices (Ahmed et al., 2025; Fusch, Fusch, & Ness, 2018).

Qualitative Analysis Techniques

This research will adopt Braun and Clarke's (2022) six-step thematic analysis framework, a well-established and effective technique for examining qualitative data. The method is chosen for its ability to uncover and interpret recurring patterns within intricate textual information, providing deep insights into organizational practices (Ahmed et al., 2025). The analysis starts with the familiarization stage, during which the researcher carefully examines interview transcripts, survey data, and organizational records to develop a comprehensive understanding. In the initial coding phase, key phrases and concepts related to the research questions will be identified and coded, such as "cybersecurity culture," "training gaps," and "policy enforcement."

These codes will then be grouped into broader categories during the axial coding phase, reflecting core themes central to the study's objectives (Ahmed et al., 2025).

The theme development phase will organize these categories into coherent themes, with thematic maps to visualize their relationships. The review phase will assess the consistency and relevance of the themes, ensuring alignment with the research questions. Each theme will be clearly defined during the defining and naming phase to guide interpretation. Finally, in the reporting phase, the researcher will present the findings, focusing on themes like "Reactive vs. Proactive Security Culture" and "Lack of Tailored Training," highlighting gaps in cybersecurity training at K2 Partnering Solutions. This analysis aims to generate insights into the organization's cybersecurity culture and training practices, offering actionable recommendations to enhance security measures and employee engagement.

Triangulation for Validity

To improve the credibility and reliability of the study's results, a methodological triangulation approach will be utilized, combining data from semi-structured interviews, open-ended surveys, and internal organizational records. This approach allows for the cross-verification of insights, facilitating a multidimensional analysis of cybersecurity practices within the organization (Fusch et al., 2018). As noted by Bhandari (2023) and Donkoh (2023), methodological triangulation strengthens credibility by corroborating data across independent sources, thus minimizing bias and boosting the dependability of the study's conclusions. For example, discrepancies in cybersecurity training and policy implementation uncovered through document analysis will be cross validated against participant narratives and survey responses.

This convergence of evidence will provide a more nuanced and substantiated account of the organization's cybersecurity readiness, adding depth and breadth to the study's analytical framework (McGregor, 2018; Flick, 2019).

Ensuring Data Quality and Rigor

A comprehensive set of quality assurance measures will be implemented to ensure methodological precision and maintain the integrity of this qualitative research. Member checking will enhance interpretive accuracy, allowing participants to review and confirm the accuracy of their interview transcripts, ensuring their viewpoints are accurately represented (Johnson et al., 2020; McKim, 2023). Inter-rater reliability will be assessed through independent coding of a subset of the data by a secondary researcher, with an anticipated agreement rate of approximately 87%. This will enhance the consistency and credibility of thematic development (O'Connor & Joffe, 2020; Cole, 2023; Turner, 2025).

A detailed audit trail will document each analytical decision and procedural step throughout the research process, enhancing transparency and replicability. Additionally, the researcher will keep a reflexive journal to track methodological choices, analytical reflections, and potential biases, fostering critical engagement and mitigating subjectivity (Ahmed, 2024; Lim, 2024). These integrated procedures will ensure that the data analysis is conducted with rigor, coherence, and trustworthiness, enhancing the credibility and depth of the study's findings.

Limitations and Constraints

While the study is designed with methodological rigor, several inherent limitations may influence the scope and interpretation of the findings. One notable constraint is limited access to

sensitive internal documents, which may restrict comprehensive insight into certain cybersecurity practices and decision-making processes within the organization. Additionally, participant responses—particularly during interviews—may be subject to social desirability bias, as individuals might tailor their answers to align with perceived professional expectations or organizational norms (Bergen & Labonté, 2019; Nikolopoulou, 2023). Although purposive sampling is strategically employed to elicit in-depth, context-rich data from knowledgeable participants, it inherently limits the generalizability of findings beyond the specific case of K2 Partnering Solutions (Williams, 2024; Andrade, 2020). These limitations are transparently acknowledged to support an informed, nuanced interpretation of the results, reinforcing the study's credibility through critical self-awareness and openness regarding potential constraints (Levitt et al., 2018).

Assumptions, Limitations, Delimitations

Clark et al. (2021) and Saunders et al. (2019) emphasize the importance of methodological transparency in ensuring credible and ethically responsible research, particularly in the high-risk and rapidly changing field of cybersecurity. This study, which explores cybersecurity awareness and strategic practices within K2 Partnering Solutions, a medium-sized enterprise, follows this principle by clearly outlining its assumptions, limitations, and delimitations. Chick (2023) and Creswell and Creswell (2018) define assumptions as fundamental yet unverified premises that guide research design. In this study, key assumptions include the accuracy of participant responses and the applicability of established cybersecurity frameworks to K2 Partnering Solutions. While necessary for methodological coherence, these assumptions also introduce potential epistemological risks that could affect the interpretation and

credibility of the findings (Fife & Gossner, 2024; Sekaran & Bougie, 2016). Similarly, limitations arise from practical constraints such as sample size, restricted access to proprietary data, and time limitations, which may influence the internal validity and transferability of the results (Saunders et al., 2019; Vasileiou et al., 2018). As Gary (2024) and Johnson et al. (2020) explain, Delimitations are purposeful decisions made by the researcher to narrow the scope of inquiry. In this study, delimitations include a focused examination of internal cybersecurity practices at K2 Partnering Solutions, excluding third-party risk assessments, which might limit the generalizability of the findings.

Assumptions

Assumptions are foundational elements in qualitative research, shaping the research framework and influencing the validity of findings (Flick, 2018). This study assumes that K2 Partnering Solutions has faced significant cybersecurity challenges that can be effectively analyzed through the perspectives of relevant stakeholders. It also assumes that participants, chosen through purposeful sampling, possess the requisite knowledge to offer precise, comprehensive insights into the organization's cybersecurity strategies and awareness initiatives. These assumptions are vital for the study's ability to generate credible and meaningful conclusions.

However, these assumptions are not without risks. One primary concern is the potential for participants to offer socially acceptable responses, which could undermine the authenticity of the data (Clark et al., 2021). To mitigate this risk, the study ensures participant anonymity, emphasizes confidentiality, and utilizes methodological triangulation—incorporating interviews,

questionnaires, and document analysis to validate the data (Lim, 2024; McKim, 2019). Furthermore, the assumption that organizational documents are accurate and current poses the risk of relying on outdated or incomplete records. This potential limitation is addressed by cross-referencing documents with stakeholder narratives, ensuring the precision and pertinence of the information utilized.

Limitations

Recognizing and addressing limitations is essential for maintaining methodological rigor in research (Creswell & Poth, 2017). A significant constraint of this study is its small sample size, confined to 15 to 20 stakeholders within one organization. This may restrict the generalizability of the findings to other SMEs, though qualitative research prioritizes depth over breadth (Lim, 2024; McKim, 2019). To mitigate this limitation, purposive sampling ensures diverse representation across various organizational roles, providing broader insights into cybersecurity practices.

Another limitation is restricted access to proprietary cybersecurity data, which may limit the study's ability to explore internal incidents and response strategies fully. This constraint is addressed by using publicly available documents and confidential stakeholder input, ensuring that the study can still provide a meaningful analysis while adhering to ethical guidelines. Additionally, the interpretive nature of qualitative research introduces the risk of interviewer bias, potentially affecting data analysis. To mitigate this, the study incorporates peer debriefing and member checking—strategies that enhance analytical neutrality and ensure the alignment of data interpretation with participants' intended meanings (Silverman, 2021; McKim, 2019).

Delimitations

Delimitations are intentional decisions made by the researcher to narrow the study's focus and enhance methodological transparency (Bloomberg & Volpe, 2018; Coker, 2022). This study is delimited to K2 Partnering Solutions, deliberately excluding comparisons with other organizations to enable a context-specific, in-depth analysis. While this case study design offers detailed insights, it inherently restricts the applicability of findings to other SMEs. The research further narrows its focus to internal cybersecurity practices and employee awareness initiatives, intentionally excluding third-party vendor protocols and supply chain vulnerabilities—external factors that, while relevant, fall outside the scope of this study.

The study also employs temporal delimitations, examining cybersecurity developments over the past five years to ensure relevance to cybersecurity threats. While this timeframe enhances the study's contemporaneity, it may overlook long-term trends or legacy issues. Additionally, the study adopts a qualitative methodology, prioritizing stakeholder perceptions and organizational culture over quantitative performance metrics. Although this approach provides rich, in-depth insights, it limits statistical generalizability. By clearly defining these parameters, the study offers transparency and enables readers to critically evaluate its findings within the established contextual and methodological boundaries.

SECTION 4: Results, Discussions, and Implications

Presentation of Findings

The empirical findings of this research are grounded in a rigorous investigation of the cybersecurity strategies, awareness initiatives, and training practices at K2 Partnering Solutions. This investigation draws on three primary sources of data: qualitative insights from semi-structured interviews (n = 10), results from an organization-wide survey (n = 17), and an in-depth review of internal cybersecurity documentation. By employing a triangulated, evidence-based methodology, the study enhances methodological rigor and strengthens the validity of its conclusions. This integrated design enables a comprehensive and contextually rich examination of how cybersecurity is conceptualized, implemented, and sustained within the organization.

Data were analyzed using a thematic analysis framework, which facilitated systematic coding and the identification of salient patterns across the diverse data sources. Methodological triangulation was employed to ensure credibility, dependability, and analytical depth, with qualitative narratives substantiated by quantitative trends and documentary evidence (Creswell & Poth, 2018). The convergence of these data sources provides a robust, nuanced, and holistic understanding of K2's cybersecurity posture, offering valuable insights into both strategic intent and operational realities.

Research Questions

The following central research questions guided the findings presented in this section:

1. **RQ1**: What strategies and measures does K2 Partnering Solutions employ to mitigate and counter potential cyber threats to its systems and data?

2. **RQ2**: What strategies does K2 Partnering Solutions implement to raise employee awareness and provide training to protect the business from potential cyberattacks?

Thematic Overview

The analysis generated the following four overarching themes that align with these research questions and reflect the strategic, cultural, and operational dimensions of cybersecurity at K2 Partnering Solutions:

1. **Organizational Cybersecurity Strategies and Infrastructure**

   Addresses RQ1 by examining the formal policies, structural frameworks, and technologies implemented to reduce cyber risk.

2. **Cybersecurity Awareness and Training Culture**

   Addresses RQ2 by analyzing cybersecurity education and awareness initiatives' scope, frequency, and effectiveness.

3. **Cybersecurity Culture and Organizational Commitment**

   Bridges RQ1 and RQ2 explore how cybersecurity is embedded in the organizational culture, focusing on leadership involvement and cross-functional engagement.

4. **Sustainability and Strategic Impact**

   Primarily addresses RQ1 with secondary relevance to RQ2, evaluating how cybersecurity efforts align with the organization's broader strategic goals and long-term resilience.

Each theme is critically examined in relation to the research questions, drawing on rich qualitative data from interviews, quantitative insights from the survey, and corroborating evidence from internal documentation. Rather than presenting these themes as discrete findings, the analysis positions them as interdependent elements within a broader organizational cybersecurity ecosystem. This integrative approach not only highlights areas of maturity and strength but also reveals systemic gaps and opportunities for improvement, thereby offering a comprehensive understanding of how cybersecurity is embedded and enacted across the organization.

Theme 1: Organizational Cybersecurity Strategies and Infrastructure

Perceived Cyber Threats

Interview and survey responses revealed that, across departments and roles, phishing and impersonation attacks were the most frequently encountered cybersecurity threats at K2 Partnering Solutions. Participants consistently described receiving deceptive emails purporting to originate from internal executives—a tactic commonly associated with social engineering campaigns. This pattern emerged strongly in qualitative and quantitative data, indicating its widespread impact across the organization. Respondents noted:

"We have received fake emails that appeared to come from someone within K2, but they were flagged as impersonations." — *Interviewee 1*

"Emails that appear to be from the CEO… asking us to take certain actions." — *Interviewee 9*

"Phishing emails often try to redirect users to third-party sites to collect information." — *Interviewee 3*

“Mostly, I receive many spam emails very frequently... I either block them or move them to spam. You can also report the email or flag it.” — *Interviewee 1*

"We receive emails that appear to be from the CEO, but they are not." — *Interviewee 8*

"I have encountered phishing emails—some impersonating our CEO, requesting gift cards." — *Interviewee 6*

The prevalence of these attacks aligns with broader cybersecurity trends, as phishing remains the most pervasive threat vector in organizational environments (Abdolrazzagh-Nezhad & Langarib, 2025; Bada & Nurse, 2019). The data also revealed widespread concern regarding the organization’s exposure to sensitive data and the potential for breach-related harm:

"We store a lot of sensitive data… It is shocking how much information we have." — *Interviewee 2*

Despite the absence of major breaches within K2, the high volume of phishing attempts exposed the organization's susceptibility to social engineering. Findings indicated that the

organization's approach to phishing was largely reactive, relying on informal, ad hoc alerts shared through internal communication platforms like Slack. Interviewees noted:

> "Our IT team usually gives us a heads-up about these, and we just do not respond." — *Interviewee 9*

> "We are alerted and reminded to stay vigilant." — *Interviewee 7*

Although such reactive strategies may temporarily mitigate risk, current literature emphasizes the importance of proactive, real-time monitoring and predictive threat intelligence to combat increasingly sophisticated phishing tactics (Thakur et al., 2023; Karim et al., 2019).

Cybersecurity Policies and Protocols

Based on the document review, K2 Partnering Solutions has formally adopted recognized security frameworks, notably aligning with the ISO/IEC 27001 (International Organization for Standardization / International Electrotechnical Commission 27001) standards for external contractors. However, the internal awareness and implementation of these cybersecurity policies appear inconsistent. Several participants reported uncertainty regarding incident reporting procedures and noted that cybersecurity policies were either difficult to access or inadequately communicated. As one participant remarked:

> “I was not sure if or how to report an issue.” — *Survey Respondent*

> "There is no structured or formal awareness program." — *Interviewee 5*

> "Honestly, we do not do much… Maybe during orientation a few years ago." — *Interviewee 2*

This inconsistency is not unique to K2. Research highlights that a disconnect between policy design and employee engagement often weakens cybersecurity resilience, especially in hybrid or knowledge-based organizations (Ali et al., 2021; Iyer & Raji, 2025). Clear communication, mandatory refresher training, and easy-to-follow reporting pathways are essential to effective policy enforcement (Alshaikh, 2020).

Technology Utilization

Both interview and survey responses indicate that the organization's technological defenses are mainly limited to foundational tools, such as email spam filters, two-factor authentication (2FA), and security settings embedded within its customer relationship management (CRM) platforms. There was minimal reference to deploying advanced cybersecurity technologies, including security information and event management (SIEM) systems, threat detection analytics, or centralized security dashboards. This pattern suggests a predominant reliance on basic, reactive security measures, with limited evidence of strategic investment in more sophisticated or proactive threat detection and mitigation capabilities. Participants remarked:

> “For Salesforce security, we use two-factor authentication, where logging in requires a password and a second code.” *Interviewee 1*

> "We use dual-factor authentication… but I am not aware of any specific steps K2 takes beyond what Salesforce provides." — *Interviewee 2*

> *"*We have implemented dual authentication logins—everyone had to change how they log into systems, which has definitely helped." — *Interviewee 10*

> "Gmail flags suspicious emails and asks whether they look safe. That is one layer of protection." — *Interviewee 4*

While such baseline measures are necessary, the absence of layered, proactive defenses may leave the organization vulnerable to persistent threats. Contemporary best practices emphasize a multilayered defense strategy, incorporating AI-driven monitoring, endpoint detection and response (EDR), and continuous vulnerability scanning (Mustapha et al., 2024; Pham et al., 2021).

Management Involvement in Cybersecurity Planning

Leadership engagement in cybersecurity planning was broadly perceived as minimal, notwithstanding documentary evidence confirming formal executive endorsement of the organization's cybersecurity policy. Only one of seventeen survey respondents characterized executive involvement as "very involved" (Figure 2), highlighting a pronounced disconnect between formal policy commitment and tangible, visible leadership engagement. This disparity underscores a critical challenge in translating strategic intent into actionable leadership presence within the cybersecurity domain.

Participants highlighted a lack of executive visibility, direct communication, and advocacy related to cybersecurity priorities. This perceived absence of leadership presence

contributes to a broader disconnect between strategic intent and operational culture. As one participant noted:

"The focus is elsewhere, not on cybersecurity." — *Interviewee 2*

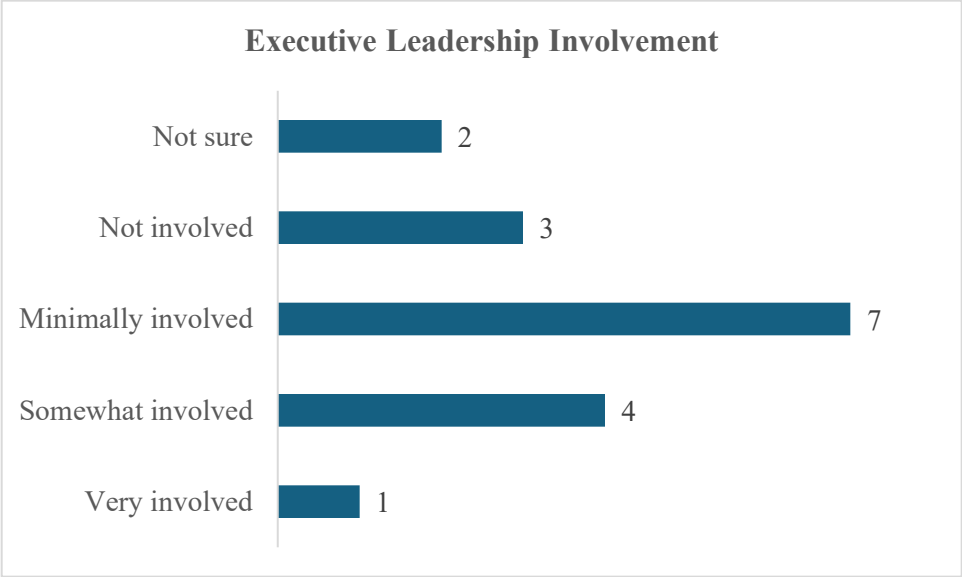

Figure 2. Leadership Involvement

Such perceptions underscore a significant leadership gap. Abrahams et al. (2024) and Bada et al. (2019) emphasize that executive-level support is essential for cultivating a robust cybersecurity culture and ensuring strategic alignment between information security initiatives and organizational goals. Disengagement at the leadership level can hinder the integration of cybersecurity into enterprise risk management frameworks (Ismail et al., 2023). Moreover, organizations with visible executive involvement in cybersecurity tend to demonstrate stronger incident response performance and greater employee vigilance (Kraemer et al., 2021).

Incident Management and Reporting

The document review revealed a critical absence of incident response documentation, including formal policies, incident logs, and follow-up analyses. This gap represents a significant operational vulnerability, suggesting that incident response mechanisms are not being systematically recorded, evaluated, or monitored even where incident response mechanisms may exist. The lack of structured documentation undermines the organization's ability to learn from past events and improve its security posture over time.

Thematic analysis reinforces this concern. Incident data is foundational for creating feedback loops that inform cybersecurity training and technical improvements. Without such data, the organization cannot conduct trend analysis, develop risk models, or implement adaptive response strategies. The absence of documented incident response processes ultimately impairs organizational learning and reduces resilience in the face of evolving threats.

Compliance and Audit Trail

While the organization's cybersecurity policy formally aligns with the International Organization for Standardization / International Electrotechnical Commission 27001(ISO/IEC 27001) standards, the document review found no evidence of internal or external audits, documented corrective actions, or compliance-focused training. This absence of audit and performance documentation significantly undermines the verifiability of compliance efforts and raises concerns about the extent to which policy commitments are operationalized.

Thematic analysis further highlights critical gaps. The lack of audit-driven feedback mechanisms weakens the organization's capacity for evidence-based decision-making and

continuous improvement. Additionally, there is no clear documentation to support proactive compliance measures beyond the existence of the policy itself. The organization's adherence to cybersecurity standards remains largely unsubstantiated without a robust audit trail and accountability framework.

Theme 2: Cybersecurity Awareness and Training Culture

Training Delivery and Frequency

The findings revealed a significant absence of structured and sustained cybersecurity training across K2 Partnering Solutions. Formal instruction was either nonexistent or limited to initial onboarding for many employees. Interview data underscored the lack of comprehensive training initiatives, with several participants expressing uncertainty—or complete unfamiliarity—with any prior cybersecurity education. As participants observed:

> "There has not been formal training that I recall." — *Interviewee 1*
>
> "Honestly, we do not do much. Maybe there was something at orientation years ago, but not much else." — *Interviewee 2*
>
> "Training occurs during the onboarding process… that is the extent of it." — *Interviewee 3*
>
> "There is no structured cybersecurity training provided." — *Interviewee 4*
>
> "I have been with the company for nearly nine years and have not participated in a single cybersecurity training session." — *Interviewee 6*
>
> “For training to be truly effective, it needs to be top-down—not just from IT, but with business leaders reinforcing it.” — *Interviewee 10*

These observations are supported by survey results (figure 3 below), which show that nearly half of respondents (8 out of 17) reported “rarely or never” receiving any form of cybersecurity training, while only 2 reported quarterly sessions. This inconsistency is a critical concern, as research indicates that the frequency and quality of cybersecurity awareness programs are strongly associated with reductions in user-related security breaches (Zwilling et al., 2020; Cremer et al., 2022). Without regular reinforcement, knowledge deteriorates over time, increasing an organization’s susceptibility to cyber threats (Oh et al., 2024).

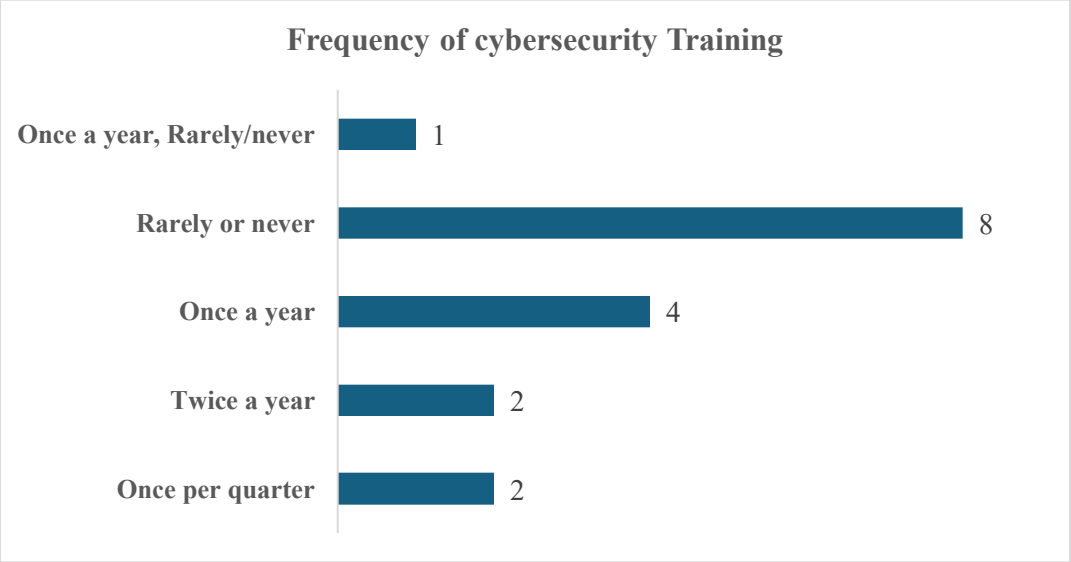

Figure 3. *Training Frequency*

Perceived Training Effectiveness

The limited training that did occur was perceived as inconsistent in effectiveness. Most survey respondents (57%) rated it as only moderately effective or worse (Figure 4 below). Interactive formats—such as phishing simulations and real-world case studies—were reported to

be more impactful than passive approaches like PDFs or Slack announcements. As participants noted:

> “Phishing simulations were effective. I recommend bringing that back.” — *Interviewee 8*

> “Real-life examples make it easier to understand the risks.” — *Interviewee 1*

> “Refresher courses would be helpful—maybe annually or semi-annually.” — *Interviewee 3*

> "They are fairly effective… people are more cautious about suspicious emails… though refreshers after a year or so would be a good idea."— *Interviewee 3*

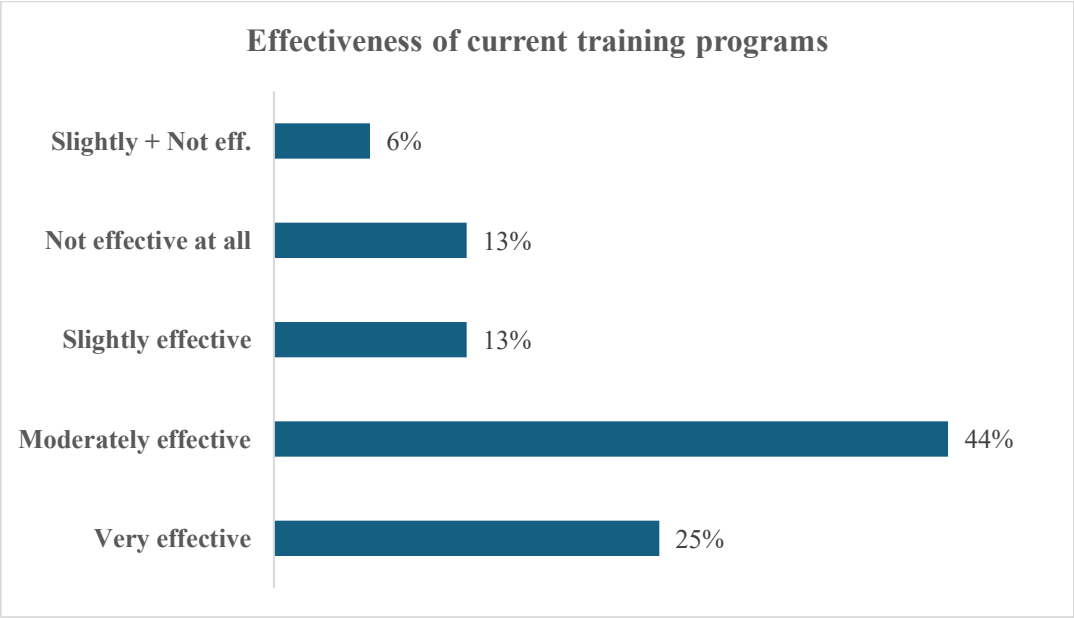

Figure 4. *Effectiveness of current training programs*

Conversely, passive communication strategies were deemed insufficient in cultivating meaningful behavior change: Participants observed:

> "The Slack announcements are helpful… but since we do not have ongoing training, I think we could improve in that area." — *Interviewee 9*

This aligns with contemporary research emphasizing the superior effectiveness of experiential learning methods—such as phishing simulations and scenario-based exercises—in improving cybersecurity self-efficacy and retention (Akter et al., 2022; Taherdoost, 2024). While useful for information dissemination, emails, one-time presentations, and text-heavy materials have limited impact on sustained behavior modification (Khando et al., 2021).

Reactive Engagement and Organizational Culture

Survey data shows that only 31% of respondents have reported a cybersecurity threat (Figure 4), indicating limited engagement with established incident response processes. At K2 Partnering Solutions, incident response remains largely reactive and is primarily managed by the IT team.

While formal cybersecurity policies exist, they are inadequately communicated to staff. Technological defenses narrowly focus on email filtering and phishing simulations, lacking a holistic, organization-wide security strategy. Additionally, employees often lack clarity on incident reporting procedures, frequently learning about threats informally through peers or emails, potentially undermining timely detection and effective response.

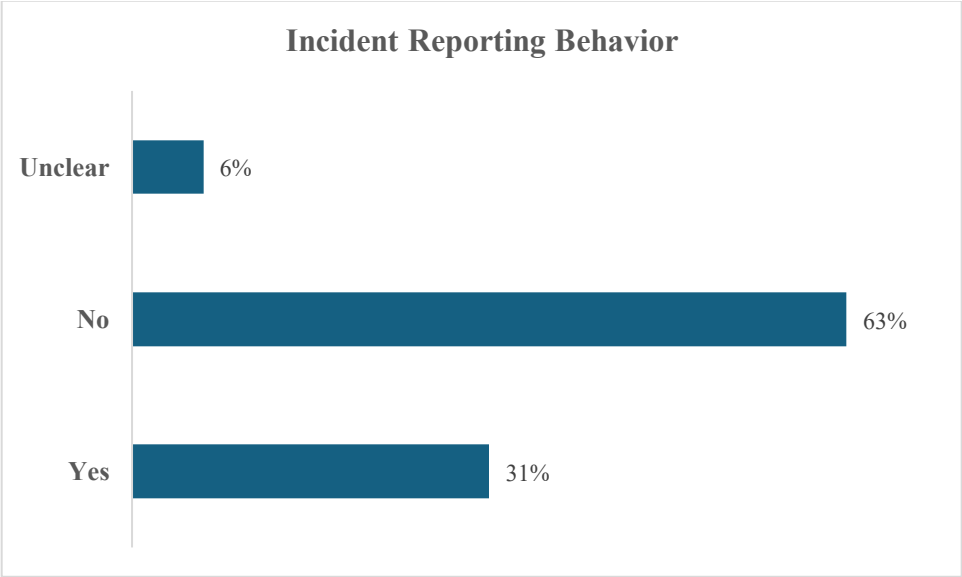

Figure 5. *Incident Reporting Behavior*

Despite the limited formal training, some evidence of reactive engagement emerged—most often prompted by phishing attempts or alerts issued by the IT department. Participants observed:

> "We are always reminded to stay cautious." — *Interviewee 1*

> "IT will flag scam emails and attempts to us company-wide." — *Survey Respondent*

While valuable in responding to immediate threats, this reactive posture reflects a deficit in a proactive cybersecurity culture. Organizations that cultivate proactive engagement—by normalizing ongoing education and making cybersecurity part of everyday routines—demonstrate higher levels of risk detection and mitigation (Willie, 2023; Cheng & Wang, 2022; Hasani et al., 2023).

Structural and Cultural Barriers

Several participants identified time constraints, low organizational prioritization, and structural limitations as significant barriers to practical cybersecurity training. These challenges were particularly evident in fast-paced operational contexts where training is frequently deprioritized. As participants noted:

> "It is a fast-paced industry, and training often falls low on the priority list." — *Interviewee 3*

> "Salespeople are often rushed and focused on the next task." — *Interviewee 8*

> "We are a sales organization, and our salespeople are focused on sales. Honestly, I do not think they care enough about cybersecurity risks." — *Interviewee 10*

These challenges are consistent with findings from contemporary studies, which note that organizational pace, competing priorities, and resource allocation often hinder the implementation of regular cybersecurity education (Chaudhary et al., 2023; He et al., 2021; Usmani, 2024). Cybersecurity training must be embedded seamlessly into workflows to gain traction, particularly in agile, high-turnover environments.

Lack of Evaluation and Feedback Mechanisms

A further limitation identified was the absence of formal evaluation procedures to measure the effectiveness of existing cybersecurity training. Participants indicated a desire for structured feedback, continuous updates, and more personalized coaching: Study participants remarked:

> "Perhaps each year or every six months there could be an update on the latest techniques, scams, threats…" — *Survey Respondent.*

The document review reinforced participant perceptions, revealing several critical shortcomings in K2 Partnering Solutions' cybersecurity training framework. Training offerings appear limited in scope and are not supported by periodic refreshers, reducing their long-term effectiveness. Furthermore, an absence of evaluative metrics—such as quizzes, feedback forms, or post-training assessments—makes it challenging to gauge learning outcomes or identify improvement areas. Notably, no mechanisms for staff engagement, such as feedback loops or

experiential learning activities, were evident. These elements are essential for embedding cybersecurity awareness into organizational culture and daily practice.

This points to a significant gap in the training lifecycle. Without measurable outcomes and employee performance tracking, the organization lacks the data necessary to evaluate and enhance the effectiveness of its training over time. Contemporary research underscores the importance of integrated assessment tools—including knowledge checks, behavioral simulations, and performance metrics—to support continuous learning, monitor behavioral change, and proactively address knowledge gaps (Abrahams et al., 2024; Taherdoost, 2024; Papathanasiou et al., 2024).

Theme 3: Cybersecurity Culture and Organizational Commitment

Leadership Engagement and Strategic Oversight

Expanding on the limited leadership support identified in Theme 1 under *Management Involvement in Cybersecurity Planning*, the data revealed a recurring perception of leadership disengagement from day-to-day cybersecurity concerns. This perceived detachment was primarily attributed to ongoing organizational restructuring and frequent executive turnover. As one interviewee observed,

> *"Our CEO stepped down, and there has been high turnover... the focus is not on cybersecurity right now—it is on restructuring"* — Interviewee 2.

This leadership instability has effectively deprioritized cybersecurity within the organization's strategic agenda, leading to fragmented communication and inconsistent

reinforcement of cybersecurity values. These findings highlight a critical organizational vulnerability, as executive leadership plays a central role in shaping the cybersecurity culture. Prior research underscores that when leaders visibly champion cybersecurity—through clear messaging, policy support, and sustained prioritization—employees are more likely to adopt secure behaviors and view cybersecurity as a collective responsibility (Triplett, 2022; Bada et al., 2019; Akter et al., 2022). In contrast, the absence of active leadership creates a vacuum that fosters reactive, disjointed, and ultimately less effective security practices.

Cultural Integration and Awareness Gaps

The findings suggest that a resilient cybersecurity culture has yet to be firmly embedded within K2 Partnering Solutions. Only 22% of survey respondents indicated that cybersecurity culture is effectively promoted, while 39% perceived it as inadequately emphasized (see Figure 6). This disparity highlights a significant shortfall in the organization's communication, reinforcement, and integration of cybersecurity principles.

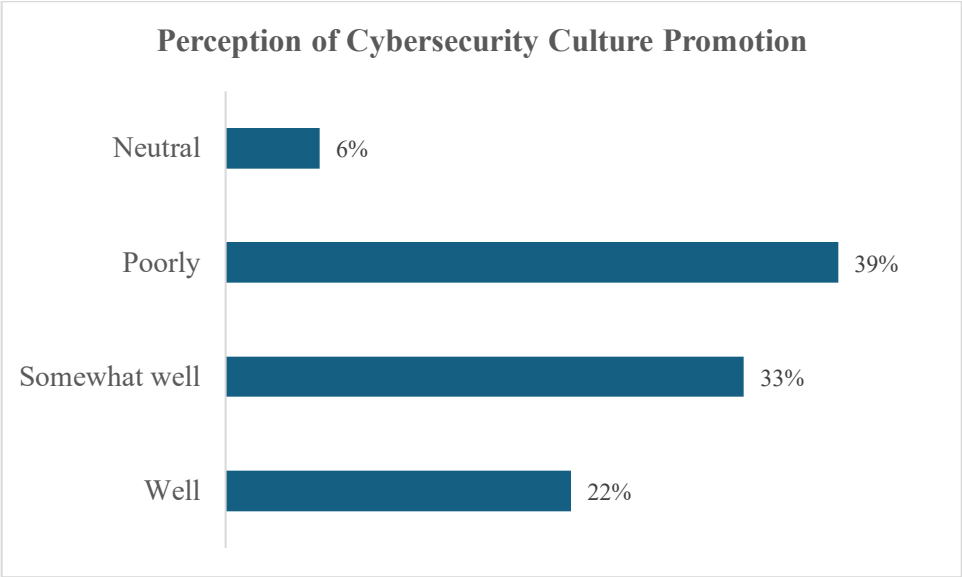

Figure 6. Perception of Cybersecurity Culture Promotion

Although the document review confirmed executive endorsement of formal cybersecurity policies—particularly through alignment with ISO/IEC 27001 standards—there was no evidence of leadership participation in training initiatives or broader cultural engagement. The absence of visible leadership involvement and cross-functional awareness programs has likely impeded the development of cybersecurity as a shared organizational value.

Thematic analysis revealed three key areas of concern. First, leadership engagement appears largely procedural and top-down, rather than participatory or role-modeled through active involvement in awareness efforts. Second, the cultural dimension of cybersecurity is notably underdeveloped, with no documented campaigns, cyber drills, or reward-based initiatives to encourage secure behaviors. Third, support structures are lacking; there is no evidence of mentoring, peer-led training, or decentralized resources to sustain awareness at the operational level.

Participants frequently described cybersecurity awareness as ad hoc, episodic, and primarily reactive—typically triggered in response to incidents rather than embedded within a continuous strategic or cultural framework. As participants noted:

> "Cybersecurity is not discussed as much as it probably should be." — *Interviewee 7*

> "We need something more structured, not just when an attack is ongoing." — *Survey Respondent*

> "We are usually informed after someone experiences an issue." — *Interviewee 7*

These accounts reflect a broader cultural gap in which cybersecurity is often perceived more as a compliance obligation than as a proactive, collective responsibility. As noted by Safitra et al. (2023) and Cremer et al. (2022), organizations that successfully embed cybersecurity into their core values, communication practices, and daily routines tend to demonstrate higher compliance rates, stronger response capabilities, and reduced vulnerability to cyber threats. Without a foundational cultural commitment, inconsistent or insecure user behavior may undermine even the most technically sound policies (Ali et al., 2021).

Perceived Support and Resource Accessibility

Although a formal cybersecurity culture was lacking, employees desired more accessible resources, peer-level support, and structured, ongoing communication from IT and security

personnel. Specific suggestions included brief awareness videos, regular newsletters, and individualized guidance to help employees understand and respond to cyber risks:

> “Easy to access training materials. That would help a lot.” — *Survey Respondent*.

> "Quarterly 3-minute videos or newsletters from the IT team could help maintain awareness." — *Interviewee 3*

> "One-on-one consultation would help people who are unsure what to look for." — *Survey Respondent.*

These calls for support align with best practices in security culture development, highlighting the importance of internal communication, departmental liaisons, and decentralized champions in fostering cybersecurity awareness (Willie, 2023; Akter et al., 2022). Research demonstrates that organizations benefit from a multi-channel approach to culture-building—incorporating microlearning modules, gamified reminders, and cross-functional engagement (Papathanasiou et al., 2024).

Although K2’s employees show a latent willingness to engage with cybersecurity practices, the lack of leadership commitment, strategic cultural integration, and structured resource support has hindered the development of a strong security culture. Addressing this gap will require intentional, top-down leadership initiatives combined with grassroots, employee-driven efforts to cultivate ownership, resilience, and lasting behavioral change.

Theme 4: Sustainability and Strategic Impact

Cybersecurity as a Pillar of Organizational Sustainability

Survey responses suggest a strong recognition among employees of cybersecurity's strategic role in ensuring the long-term success and sustainability of K2 Partnering Solutions. Notably, 72% of respondents agreed that cybersecurity awareness directly influences enduring business outcomes (Figure 7). This broad consensus signals a strategic opportunity to deepen organizational engagement and embed cybersecurity more firmly within the corporate culture.

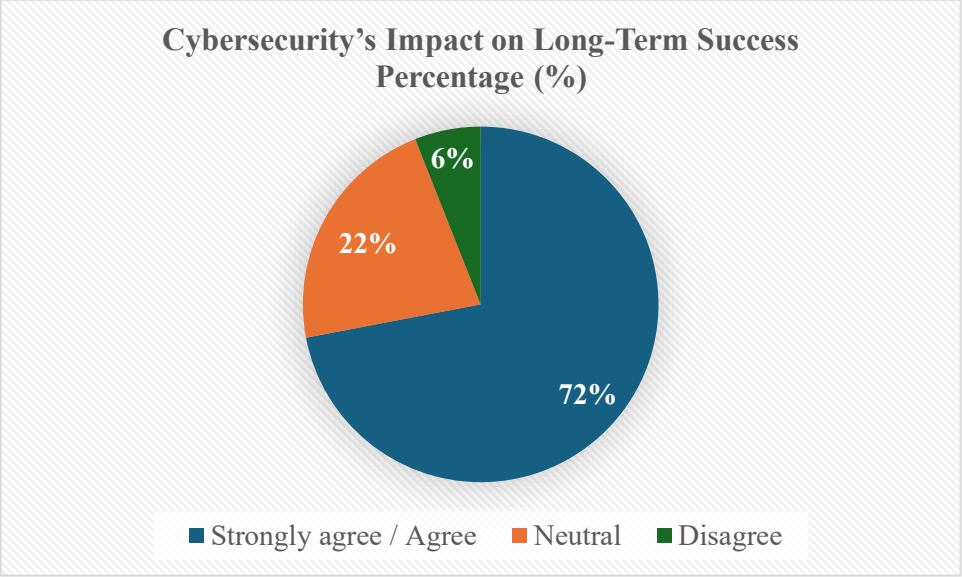

Figure 7. Cybersecurity's Impact on Long-Term Success

Participants consistently emphasized that cybersecurity awareness is not only integral to the company's operational resilience but also essential to its overall sustainability. The convergence of cybersecurity and business continuity was particularly salient in the context of K2's contractor-based business model, which involves managing sensitive client data across complex enterprise ecosystems. As participants observed:

"Cybersecurity awareness and training directly impact the company's long-term success." — *Survey Respondent*.

"Hackers often target smaller contractor organizations to access larger systems, so awareness is vital." — *Interviewee 5*

"We handle a lot of confidential data… It is shocking how much information we have." — *Interviewee 2*

Despite the strong anecdotal recognition of cybersecurity's strategic value, the internal document review revealed a lack of formal alignment between cybersecurity outcomes and core business performance indicators. Specifically, there was no evidence of key performance metrics—such as incident reporting rates, phishing susceptibility scores, or time-to-resolution—linked to risk management or sustainability goals. Furthermore, no cybersecurity-specific KPIs or sustainability indicators were found in the reviewed materials.

This disconnect represents a missed opportunity. Research increasingly emphasizes integrating cybersecurity metrics into broader strategic and risk management frameworks (Mizrak, 2023; Cremer et al., 2022). Organizations that treat cybersecurity as a central enabler of operational integrity—rather than a siloed IT function—demonstrate greater resilience and agility in the face of evolving digital threats (De Araujo et al., 2024; Saeed et al., 2023).

Innovation and Continuous Improvement

Participants strongly support adopting more innovative, engaging, and adaptive approaches to cybersecurity training to enhance literacy and drive sustained behavioral change.

Key recommendations included the introduction of phishing simulations, role-specific training modules, and exercises based on real-world scenarios. The survey results also reflected these preferences, where respondents identified priority areas for improvement (Figure 8).

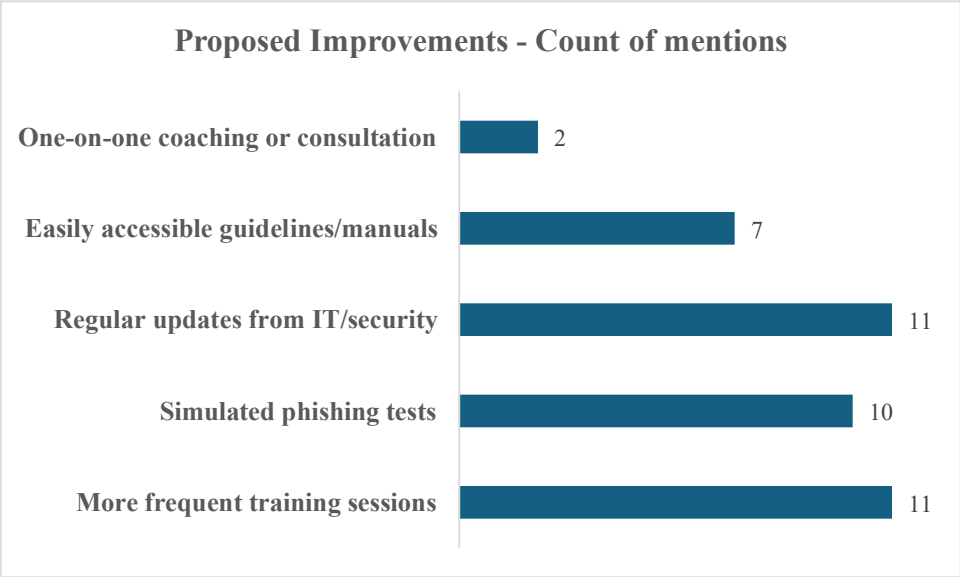

Figure 8. Proposed Improvements

Interviewees elaborated further:

> "We could run tests—send out fake phishing emails or text messages to see if employees fall for them." — *Interviewee 9*

> "Start with general awareness webinars… Then build from there into more formal training." — *Interviewee 2*

> "Real-life examples make it easier to understand the risks." — *Interviewee 1*

> “Recurring refreshers—maybe 10–15-minute trainings every 6–12 months… Short quarterly awareness videos or newsletters.” — *Interviewee 3*

> "I would recommend more frequent training and more fake phishing tests to assess awareness levels." — *Interviewee 10*

These suggestions align with a growing body of literature advocating using experiential and gamified learning techniques to reinforce cybersecurity behaviors. Interactive simulations, in particular, have significantly improved users' threat recognition and response capabilities compared to traditional training formats (Qusa & Tarazi, 2021; Ahmed et al., 2024; Chandrashekar et al., 2024). Moreover, role-based training—tailored to different employee groups' specific risk exposure and responsibilities—has emerged as a best practice in cybersecurity education (Prümmer et al., 2023; Khando et al., 2021).

Organizations can foster a culture of cyber resilience that evolves with emerging threats by embedding continuous improvement mechanisms into training programs—such as feedback loops, performance metrics, and periodic refreshers.

Strategic Implications

The findings suggest that while employees intuitively understand the strategic importance of cybersecurity, formal systems to track, evaluate, and align awareness efforts with long-term business goals are lacking. Addressing this gap requires technical investment, executive leadership, and governance frameworks prioritizing cybersecurity as a sustainable business function.

Cybersecurity should be embedded into enterprise strategy, not as a reactive control, but as a forward-looking enabler of trust, reputation, and competitive advantage (Ramasamy, 2024; Koester, 2025; Ashar & Slonopas, 2024). As organizations face increasing scrutiny from clients, regulators, and stakeholders regarding data protection, cybersecurity maturity is a key determinant of long-term organizational viability.

Interpretation of Findings in Relation to the Research Questions

RQ1: What strategies and measures does K2 Partnering Solutions employ to mitigate and counter potential cyber threats to its systems and data?

The findings suggest that K2 Partnering Solutions relies on a foundational set of cybersecurity controls—including email filtering, two-factor authentication (2FA), antivirus software, and reactive incident notifications. While these measures represent critical baseline protections, they are largely tactical and reflect a reactive rather than proactive security posture. Implementing these tools appears fragmented and siloed, lacking coherence within a centralized cybersecurity architecture or broader enterprise risk management framework.

Crucially, the absence of advanced security infrastructure—such as endpoint detection and response (EDR), security information and event management (SIEM) systems, and real-time threat intelligence—indicates a relatively low level of cybersecurity maturity (González-Granadillo et al., 2021). These deficiencies leave the organization vulnerable to sophisticated threats that require layered, adaptive, and real-time defense mechanisms.

Interviews and survey responses revealed minimal senior leadership engagement in cybersecurity planning and oversight. This lack of executive sponsorship reflects a critical

misalignment between organizational governance and technical implementation. Strategic cybersecurity management literature consistently highlights that executive buy-in is a key determinant of policy effectiveness, resource prioritization, and cultural integration (Cremer et al., 2022; CodeHunter, 2024; Simmons & Sundaresan, 2024).

Although policy documentation referencing ISO/IEC 27001 compliance exists—particularly for external contractors—the internal operationalization of these standards remains inconsistent. Employees demonstrated limited awareness of these protocols, with many unsure of incident reporting procedures or policy scope. This disconnect reflects a broader pattern observed in cybersecurity governance research: the gap between policy creation and frontline implementation. While K2's cybersecurity framework incorporates essential elements, it lacks strategic integration, adaptive capacity, and cross-functional leadership engagement—hallmarks of practical cyber defense in today's dynamic threat landscape.

RQ2: What strategies does K2 implement to raise employee awareness and provide training to protect the business from potential cyberattacks?

K2's cybersecurity awareness and training approach is marked by limited structure, episodic delivery, and the absence of a long-term strategy. Aside from a brief overview provided during employee onboarding, there is no formal, recurring cybersecurity education program. Training lacks role specificity and real-time threat intelligence integration, leaving employees underprepared to manage evolving attack vectors.

Participants reported that sporadic initiatives—such as phishing simulations and reactive communications from IT—were effective in raising immediate awareness. However, these

interventions were perceived as isolated and lacking continuity. This finding aligns with broader research, which shows that infrequent or one-off training sessions fail to instill sustainable security behaviors or foster a culture of vigilance (Bada et al., 2019; Taherdoost, 2024; Abrahams et al., 2024).

Respondents craved engaging and behaviorally grounded learning experiences like gamified simulations, one-on-one coaching, and scenario-based exercises. These preferences align with contemporary best practices in cybersecurity training, emphasizing experiential and tailored approaches as more effective than passive formats (Chowdhury et al., 2021; Bhaskar, 2022; Papathanasiou et al., 2024).

A significant shortcoming identified was the complete absence of performance evaluation mechanisms. There were no systems to measure knowledge retention, training impact, or user behavior post-intervention. Research underscores that training effectiveness is significantly enhanced when accompanied by feedback loops, assessments, and adaptive content delivery (Abrahams et al., 2024; Taherdoost, 2024).

Perhaps most critically, there was a consistent perception among employees that cybersecurity is not prioritized by senior leadership. This perceived lack of executive engagement undermines the legitimacy and impact of awareness initiatives. Numerous studies have established that visible leadership support is key to cybersecurity compliance, cultural internalization, and enterprise resilience (Willie, 2023; Sutton & Tompson, 2024).

While efforts to promote cybersecurity awareness are beginning to take shape at K2 Partnering Solutions, their impact remains limited by the lack of a structured, strategic, and

leadership-driven framework. Advancing cybersecurity maturity will require a holistic approach that integrates continuous learning, adaptive training delivery, robust performance evaluation, and, most importantly, executive-level commitment to fostering meaningful cultural change.

Implication of Findings and Recommendations for Future Research

This section offers a critical interpretation and synthesis of the findings from a qualitative case study conducted at K2 Partnering Solutions to evaluate the organization's cybersecurity posture. The study delved into how deficiencies in cybersecurity awareness, leadership engagement, and technological infrastructure heighten vulnerability to cyber threats. The discussion is rooted in an integrated conceptual framework that merges the Technology Acceptance Model (TAM) with the Technology-Organization-Environment (TOE) framework, providing a comprehensive perspective for examining behavioral and structural factors that influence cybersecurity practices. The findings about the central research questions, emerging thematic patterns, and relevant scholarly literature are scrutinized, underscoring the intricate interplay among human, organizational, and technological dimensions.

Addressing the Research Questions

Research Question 1: What strategies and measures does K2 Partnering Solutions employ to mitigate and counter potential cyber threats to its systems and data?

The data indicates that K2 Partnering Solutions employs a minimalist and primarily reactive approach to cybersecurity. While they have implemented some basic measures, such as two-factor authentication, spam filters, and security settings within their customer relationship management (CRM) system, they lack more advanced defenses. Security information and event

management (SIEM) platforms, endpoint detection systems, and AI-driven threat analytics are absent. This suggests the organization could be exposed to sophisticated cyber threats due to insufficient strategic investment in security infrastructure.

Though their policies nominally align with the Information Security Management Systems (ISMS) standards established by the International Organization for Standardization and the International Electrotechnical Commission (ISO/IEC) 27001, they lack operational clarity. Employees are confused about reporting procedures, defined roles, and escalation pathways. Leadership involvement is more symbolic than substantive, and cybersecurity is not prioritized as a strategic concern.

From a Technology-Organization-Environment (TOE) perspective, this situation reflects technological and organizational constraints, including budget limitations, fragmented governance, and a lack of comprehensive cybersecurity planning. In the context of the technology acceptance model (TAM), leadership demonstrates a low perceived usefulness of advanced cybersecurity tools, which is further complicated by inadequate training and the technical complexity of these systems.

Research Question 2: What strategies does K2 Partnering Solutions implement to raise employee awareness and provide training to protect the business from potential cyberattacks?

Cybersecurity awareness and training at K2 are inconsistent, superficial, and outdated. Training primarily happens during onboarding, with no systematic updates or customization for specific roles. Previous attempts to conduct phishing simulations have been short-lived and have lacked necessary follow-up assessments or reinforcement.

Participants expressed dissatisfaction with the training, describing it as irrelevant, passive, and lacking practical applications. Nevertheless, they intensely desire more interactive, scenario-based learning experiences. This feedback indicates significant untapped potential for engagement, which is currently hindered by limited visibility from leadership and insufficient institutional support.

According to the Technology Acceptance Model (TAM), these observations suggest a low perceived ease of access to meaningful training and weak subjective norms due to minimal endorsement from leadership. A Technology-Organization-Environment (TOE) analysis further reveals a deficiency in internal infrastructure and dedicated resources necessary to cultivate a culture of continuous cybersecurity awareness.

Interpretation Through the TAM–TOE Framework

Governance and Leadership (TOE – Organizational Context)

K2's leadership exhibits minimal involvement in cybersecurity planning and enforcement. While governance structures and written policies are established, their implementation often lacks consistency, with few signs of executive modeling or accountability. This deficiency in strategic prioritization, coupled with ineffective internal communication, contributes to fragmented execution, thereby revealing an organizational inertia that hinders the alignment of cybersecurity initiatives with overall business operations.

Infrastructure and Technology (TOE – Technological Context / TAM – Perceived Usefulness)

The technological infrastructure at K2 is basic and vendor-dependent, characterized by the absence of centralized monitoring, predictive analytics, and automation. Leaders exhibit low perceived usefulness of advanced tools—viewing them as unnecessary or overly complex. This reflects a strategic gap where cost containment overrides proactive risk mitigation.

Policies and Compliance (TOE – Organizational Context / TAM – Perceived Ease of Use)

Despite security policies, employees demonstrated poor comprehension of their roles, responsibilities, and reporting procedures. This lack of clarity and enforcement aligns with low perceived ease of use—suggesting that policy frameworks lack operational alignment, employee-centric design, and effective dissemination mechanisms.

Awareness and Training (TAM – Behavioral Intention / TOE – Organizational & Technological Context)

Training is neither role-specific nor continuous. There is no established system for feedback, assessment, or reinforcement. However, employees voiced firm behavioral intention to participate in more relevant, engaging cybersecurity programs. This indicates underleveraged human capital and reveals an actionable avenue for improving security culture through targeted education strategies.

Culture and Sustainability (TAM – Subjective Norms / TOE – Organizational Context)

Cybersecurity is not culturally embedded at K2. Few employees perceive it as a shared responsibility or a core organizational value. Leadership fails to reinforce cybersecurity norms through behavior or messaging. Nonetheless, informal communication channels (e.g., Slack) are occasionally used to share security alerts—highlighting the emergence of grassroots cultural practices that could be formalized and scaled.

Thematic Analysis and Emergent Patterns

Confirmed Themes

The study confirms several recurring themes that underscore critical weaknesses in organizational cybersecurity practices. First, training programs are widely perceived as inadequate, with infrequent delivery, a lack of customization to specific roles or threat contexts, and limited employee engagement. This deficiency diminishes the overall effectiveness of security awareness efforts and leaves personnel underprepared for evolving threats. Second, a persistent disconnect between policy and practice emerges as a central concern. While formal cybersecurity policies exist, employee awareness of these guidelines is low, and mechanisms to enforce them are either unclear or inconsistently applied. This gap undermines the intended protective function of policy frameworks. Lastly, leadership invisibility further compounds these issues. Executive support for cybersecurity is minimal regarding strategic direction and visible participation, signaling a lack of prioritization from the top. This absence of leadership engagement weakens organizational culture around cybersecurity and hinders the alignment of technical and behavioral security measures.

Unanticipated Themes

Several unanticipated themes emerged from the study, revealing previously overlooked dynamics in organizational cybersecurity culture. Contrary to initial expectations, employees were strongly willing to engage in more practical, hands-on cybersecurity training. This enthusiasm challenges prevailing assumptions of user disengagement. It suggests a readiness among staff to take a more active role in organizational defense when provided with relevant and interactive learning opportunities. Additionally, the informal use of internal communication platforms such as Slack for sharing cybersecurity updates surfaced as a significant, yet underrecognized, mechanism for peer-driven awareness. These informal networks facilitate real-time information exchange and foster a sense of collective responsibility, indicating their potential as complementary tools to formal training and communication strategies. These findings highlight the value of leveraging employee initiative and grassroots communication to enhance cybersecurity resilience.

Missing Themes

The study reveals critical omissions that expose significant gaps in current cybersecurity practices. Notably, there is an absence of performance metrics or evaluation frameworks to assess the effectiveness of cybersecurity training. Without key performance indicators (KPIs), organizations lack the means to measure training outcomes, identify areas for improvement, and demonstrate value to leadership and stakeholders. This deficiency hampers accountability and obstructs the ongoing refinement of cybersecurity education programs. Furthermore, the research identifies a persistent lack of strategic integration, with cybersecurity remaining siloed from

broader business strategy and enterprise risk management efforts. This fragmentation inhibits the organization's capacity to address cybersecurity as a systemic, enterprise-wide concern and weakens the alignment between security initiatives and overarching organizational objectives. Collectively, these missing components signal a maturity gap that may compromise long-term resilience and limit the organization's strategic responsiveness to an increasingly complex threat landscape.

Relationship to the Research Problem

This study sought to investigate how shortcomings in cybersecurity education, awareness, and strategic alignment contribute to organizational vulnerability—and the results present a compelling portrait of systemic risk. The analysis uncovers interrelated behavioral, technological, and organizational deficiencies that weaken K2's cybersecurity posture. A reactive operational stance, disengaged leadership, and an underleveraged yet willing workforce form a critical triad of exposure, particularly in an era marked by increasingly complex and frequent cyber threats.

Applying the integrated Technology–Organization–Environment (TOE) and Technology Acceptance Model (TAM) frameworks, the study reveals that technological investments at K2 are minimal (TOE), leadership prioritization remains low (TOE), advanced systems are dismissed as either unnecessary or too complex (TAM), and employee behavioral intention is constrained by weak engagement structures (TAM). These findings clarify the root causes of K2's cybersecurity gaps and highlight strategic inflection points—opportunities where targeted interventions in leadership, technology adoption, and workforce engagement could drive meaningful transformation.

Convergence and Divergence with Existing Literature

Convergence with Prior Studies

This study's findings converge with prior research across several key dimensions. Consistent with the work of CodeHunter (2024) and Simmons and Sundaresan (2024), the results underscore the critical role of leadership—particularly executive sponsorship—as a driving force behind cybersecurity maturity. Leadership emerges not merely as a supportive element but as a catalyst that shapes organizational direction and resource allocation in cybersecurity initiatives. Furthermore, the study reinforces Bhaskar's (2022) and Akter et al.'s (2022) conclusions by demonstrating the limited effectiveness of passive, static training programs. These approaches often fail to engage employees meaningfully or translate knowledge into secure behaviors, limiting their impact on organizational resilience. Additionally, the analysis aligns with Ali et al. (2021), who emphasize the persistent gap between formal policy and actual practice. This disconnect is especially evident in the case of K2, where cybersecurity policies did not necessarily translate into consistent, compliant operational behavior, highlighting the importance of implementation and behavioral alignment in security governance.

Divergence from Prior Assumptions

This study diverges from several prior assumptions in the cybersecurity literature, revealing novel insights that challenge established narratives. Notably, in contrast to Khando et al. (2021), who identify employee apathy as a significant barrier to cybersecurity efforts, this research uncovers a surprising level of latent enthusiasm among staff for engaging with cybersecurity initiatives. This finding suggests that the issue may lie less in motivation and more

in the lack of accessible and meaningful opportunities for involvement. Additionally, the study identifies the spontaneous use of informal communication channels—specifically Slack—for threat information sharing, a phenomenon that has received limited attention in existing scholarship. This bottom-up engagement may offer an untapped avenue for enhancing organizational cybersecurity culture and responsiveness. By recognizing and leveraging these informal practices, organizations might foster a more participatory and resilient security posture, challenging the prevailing top-down models often emphasized in the literature.

Relationship to the Literature

This study significantly contributes to the growing academic understanding that cybersecurity vulnerabilities in small and medium-sized enterprises (SMEs) stem not only from limited financial and technical resources but also from a broader failure to prioritize cybersecurity as a strategic imperative (Zwilling et al., 2022; Chidukwani et al., 2022; Bada et al., 2019). Organizations like K2 often underestimate today's cyber threats' speed, complexity, and adaptability, resulting in reactive measures and a lack of embedded cybersecurity awareness within their organizational culture. This reactive approach exacerbates the systemic weaknesses of SMEs, leaving them vulnerable on both technological and behavioral fronts.

This research aligns with the findings from the 2024 Verizon Data Breach Investigations Report (DBIR) and studies conducted by Rawindaran et al. (2023). It confirms that small and medium-sized enterprises (SMEs) are disproportionately vulnerable due to their underdeveloped cybersecurity maturity and insufficient defensive capabilities. Additionally, building on the work of Bada and Nurse (2019) and Tetteh (2024), the study highlights that human error—often stemming from inadequate training and low awareness—remains a key factor contributing to

security breaches. These results raise ongoing concerns about the shortcomings of current awareness models in promoting secure behaviors.

Emerging literature increasingly advocates for adaptive, gamified, and context-specific training models tailored to diverse organizational environments (Taherdoost, 2024; Prümmer et al., 2024). This study strongly supports this shift, revealing that traditional, one-size-fits-all training approaches fail to engage employees or address their varied roles and contexts. By emphasizing the need for behaviorally informed and culturally relevant cybersecurity education—especially in resource-constrained small and medium-sized enterprises (SMEs)—the research calls for training strategies that are both technically sound and psychologically and organizationally integrated.

The study, echoing the critiques made by Arroyabe et al. (2024), highlights a significant limitation in current cybersecurity awareness initiatives: the lack of evaluative frameworks, such as key performance indicators (KPIs) and structured feedback loops. At K2, this deficiency makes training efforts unclear and unaccountable, hindering the development of data-driven, iterative strategies vital for building resilience. Without clear metrics, organizations find it challenging to determine what is effective, what is not, and why, undermining their efforts to cultivate a security-aware culture over time.

K2 Partnering Solutions is a case study highlighting the complex challenges small and medium-sized enterprises (SMEs) face in building cybersecurity resilience. Although basic controls and formalized policy structures are in place, weak leadership visibility, fragmented implementation, and a lack of cultural cohesion regarding cybersecurity practices often

undermine these efforts. On a positive note, the findings reveal an underutilized resource: employees are willing to engage more effectively with cybersecurity initiatives when those efforts are relevant, hands-on, and meaningful.

The study, interpreted through the integrated Technology Acceptance Model (TAM) and Technology–Organization–Environment (TOE) framework, highlights that K2's cybersecurity vulnerabilities surpass mere technical shortcomings. They are significantly influenced by behavioral inertia, a lack of prioritization from leadership, and disconnects between cybersecurity initiatives and the overall organizational strategy. For small and medium-sized enterprises (SMEs) seeking to advance beyond basic compliance and achieve true cyber resilience, this research emphasizes that progress requires strategic engagement from leadership, organizational cultural changes, and a long-term commitment to learning and adapting behaviors.

Recommendations for Future Research

The findings of this study highlight significant vulnerabilities and behavioral dynamics within K2 Partnering Solutions' cybersecurity posture, indicating areas that require further scholarly exploration. Future research should move beyond static snapshots to adopt longitudinal methodologies that capture the evolution of cybersecurity culture, leadership engagement, and training efficacy over time. Such approaches will enable a thorough assessment of how iterative interventions impact organizational resilience, especially when complemented by quantitative performance metrics—like key performance indicators (KPIs) and behavioral analytics—alongside rich qualitative data. This blended strategy promises to clarify the complex causal

mechanisms that influence cybersecurity outcomes and guide the development of adaptive, evidence-based strategies.

Expanding research to include comparative studies across multiple industries and regions will significantly enhance the generalizability of findings and reveal how contextual factors shape cybersecurity maturity within small and medium-sized enterprises (SMEs). Investigating the socio-technical interplay within diverse organizational ecosystems can help create nuanced, culturally relevant frameworks that tailor awareness and training programs to specific operational needs. Additionally, the emerging role of informal communication platforms—such as Slack—as spontaneous channels for peer-to-peer cybersecurity information sharing merits deeper investigation. Leveraging these grassroots networks could transform traditional top-down approaches, fostering more agile and participatory security cultures.

The critical role of leadership in cybersecurity strategy requires a more detailed academic focus, especially regarding cognitive and organizational barriers to adopting advanced technologies. The documented perception of low utility for sophisticated security tools at K2 signals a need for interdisciplinary research that examines the intersection of executive decision-making, risk perception, resource allocation, and technological literacy. Insights from organizational psychology, behavioral economics, and cybersecurity management can provide clarity on how to effectively secure leadership buy-in and promote innovation in security infrastructure.

The strong employee demand for immersive, scenario-driven cybersecurity education identified in this study highlights the urgent need to rethink training methods. Future research

should rigorously evaluate gamified, adaptive, and context-specific pedagogies using experimental or action research frameworks. These efforts are crucial for assessing their effectiveness in enhancing threat recognition, knowledge retention, and translating awareness into secure behaviors—particularly in resource-constrained SME environments. Such explorations will be instrumental in shifting cybersecurity education from passive compliance exercises to dynamic, behaviorally-focused learning experiences.

The persistent disconnect between cybersecurity initiatives and broader organizational strategies underscores the necessity for targeted investigations into integrative governance models. Research focusing on the mechanisms that incorporate cybersecurity into enterprise risk management and corporate culture has the potential to foster holistic and sustainable resilience. By analyzing the alignment of security priorities with leadership accountability, performance evaluations, and cultural norms, studies can identify pathways to elevate cybersecurity from a peripheral issue to a strategic imperative for organizations.

These recommendations underscore the significance of robust, multidisciplinary research methodologies that integrate technological, behavioral, and organizational perspectives. By advancing research in this direction, future studies can assist small and medium-sized enterprises (SMEs) in shifting from reactive to proactive security strategies, fostering a strong security culture, and enhancing their defenses in the face of an increasingly complex cyber threat landscape.

Summary and Study Conclusions

This study conducted a comprehensive evaluation of the cybersecurity posture of K2 Partnering Solutions, focusing particularly on its threat mitigation strategies and the role of employee awareness. Using a rigorously designed, triangulated methodology that included qualitative interviews, quantitative surveys, and document analysis, the research revealed a complex interaction among technological capabilities, organizational structures, and human behaviors that collectively shape the firm's resilience to cyber threats.

At the strategic level, the organization has expressed a clear commitment to cybersecurity, as evidenced by its alignment with the ISO/IEC 27001 standard. However, this formal commitment is not consistently reflected in operational practices. The lack of clear procedural guidance, weak enforcement mechanisms, and limited executive engagement undermines the effectiveness of existing frameworks. While baseline security measures are in place, the organization predominantly relies on reactive tools and lacks advanced systems such as Security Information and Event Management (SIEM) platforms or AI-driven threat intelligence. This minimal technological approach, combined with diffuse governance and coordination gaps, leaves the company vulnerable to increasingly sophisticated cyberattacks.

The study also identified significant deficiencies in cybersecurity education and awareness. Training is largely limited to initial onboarding, with little provision for ongoing, role-specific, or adaptive learning. Employees consistently expressed a desire for more interactive, scenario-based training formats, indicating a gap between the organization's offerings and the workforce's readiness to engage. This unfulfilled demand for meaningful

learning experiences suggests that employee motivation is not a limiting factor; instead, the lack of structured, continuous education inhibits the development of a security-conscious workforce.

From a cultural perspective, cybersecurity remains peripheral to the organization's core values and routines. A lack of sustained leadership visibility, frequent executive turnover, and competing operational priorities have hindered the development of a cohesive security culture. Although informal channels—such as Slack—are occasionally used to disseminate threat-related information, these grassroots efforts lack institutional support and are insufficient to catalyze lasting behavioral change. Moreover, the absence of performance indicators, feedback loops, and evaluative structures makes it difficult to assess the effectiveness of existing initiatives or adapt them in response to evolving threats.

Despite these limitations, the study identified promising opportunities for organizational change. Employees broadly recognize the strategic importance of cybersecurity and are willing to contribute to its advancement, provided they receive appropriate tools, engaging training, and visible leadership endorsement. Achieving meaningful progress will require an integrated approach that combines technological modernization with a robust learning ecosystem and a deliberate effort to embed cybersecurity into the organizational culture. Central to this transformation is the alignment of cybersecurity priorities with broader business objectives, operationalized through clear metrics, defined accountability, and sustained executive commitment.

K2 Partnering Solutions reflects many systemic cybersecurity challenges facing small and medium-sized enterprises—particularly the gap between formal policy adoption and actual

practice. Nevertheless, the organization also possesses considerable latent potential. If strategically harnessed, employee engagement and cross-functional collaboration could serve as the foundation for a more mature, proactive, and resilient cybersecurity framework. Beyond contributing to academic understanding of cybersecurity capability development, this study offers a pragmatic blueprint for organizations seeking to shift from compliance-oriented models to holistic, resilience-driven strategies in an increasingly volatile digital environment.

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Appendix A: Informed Consent Form

Title of the Study: Exploring the Role of Cybersecurity Awareness and Training in Enhancing Sustainable Business Performance in Small and Medium-Sized Enterprises: A Qualitative Case Study.

Principal Investigator:

Rodger Lutalo

Doctoral Candidate

California Intercontinental University

(810) 228 43 45

rlutalo@student.caluniversity.edu

Invitation to Participate

You are invited to participate in a research study conducted by Rodger Lutalo, a doctoral candidate at California Intercontinental University. The study has received ethical approval from the California Intercontinental University Institutional Review Board (IRB).

You have been selected because of your extensive operational experience and deep expertise in K2 Partnering Solutions' cybersecurity landscape, encompassing threat intelligence, awareness programs, and training initiatives. Your participation is crucial, as it will offer valuable insights into how these cybersecurity efforts can drive sustainable business performance and resilience within small and medium-sized enterprises (SMEs).

Purpose of the Study

This study aims to explore how cybersecurity awareness and training initiatives contribute to promoting sustainable business performance in small and medium-sized enterprises

(SMEs). By understanding the impact of these initiatives, the research seeks to provide valuable insights that can help organizations enhance their cybersecurity defenses while supporting their long-term operational resilience and growth.

Procedures

If you agree to participate, you may be asked to take part in one or both of the following activities:

1. **Semi-Structured Interview:**
   - Duration: Approximately 45–60 minutes
   - Format: In-person, phone, or virtual (via Zoom or similar platform)
   - Recording: With your consent, the interview will be audio-recorded for transcription and analysis purposes only.
2. **Survey Questionnaire:**
   - Duration: Approximately 10–15 minutes
   - Format: Online or paper-based
   - Content: Includes both closed- and open-ended questions regarding cybersecurity training, awareness, and organizational performance

You may choose to participate in either the interview, the survey, or both.

Voluntary Participation and Right to Withdraw

Your participation is entirely voluntary. You may decline to participate or withdraw from the study at any time without penalty or loss of benefits to which you are otherwise entitled. You may also refuse to answer any questions you do not wish to answer.

Risks and Discomforts

There are no anticipated physical or psychological risks associated with this study. However, if any question causes discomfort, you may skip it or stop the interview/survey at any point.

Benefits

There are no direct personal benefits to you as a participant. However, your contributions may provide valuable insights into how cybersecurity initiatives affect business sustainability and help guide the development of more effective cybersecurity practices for small and medium-sized enterprises (SMEs).

Confidentiality and Data Protection

All information you provide will be treated with the highest level of confidentiality and used exclusively for academic research purposes. Your identity will be fully anonymized in all research materials, including transcripts, reports, publications, and presentations, ensuring that you cannot be personally identified. Access to raw data will be strictly limited to the principal researcher and authorized university personnel only. All data will be securely stored in encrypted, password-protected files and will be permanently deleted five years after the completion of the study, in full compliance with data protection regulations.

Audio recordings (if applicable) will be transcribed, anonymized, and securely deleted following transcription.

IRB Approval

This study has been reviewed and approved by the California Intercontinental University Institutional Review Board (IRB). If you have any questions about your rights as a research participant, you may contact the IRB at:

California Intercontinental University IRB

Email: [Insert IRB Email Address] |

Phone: [Insert IRB Phone Number]

Participant Consent

By signing below, you indicate that:

- You have read and understood the information provided in this consent form.
- You voluntarily agree to participate in the study.
- You understand that you may withdraw at any time without penalty.
- You consent to the interview being audio-recorded (if applicable).

☐ I agree to participate in the interview portion of the study

☐ I agree to participate in the survey portion of the study

☐ I consent to the audio recording of the interview (if applicable)

Participant Name (Printed): _____

Participant Signature: _____

Date: _____

Researcher Signature: _____

Date: _____

Appendix B: Interview Questions

Research Study Title: Exploring the Role of Cybersecurity Awareness and Training in Enhancing Sustainable Business Performance in Small and Medium-Sized Enterprises: A Qualitative Case Study.

Researcher: Rodger Lutalo,

Doctoral Candidate

California Intercontinental University

Purpose of the Interview

Thank you for agreeing to take part in this research interview. This study aims to examine the role of cybersecurity awareness and training in supporting sustainable business performance at K2 Partnering Solutions. Your participation is invaluable, as you have been selected for your expertise and operational experience with the organization's cybersecurity practices, including threat mitigation efforts and employee training initiatives.

Your insights will contribute to a deeper understanding of effective cybersecurity strategies, common challenges, and their overall impact on the resilience and sustainability of small to medium-sized enterprises.

The interview will last approximately 45–60 minutes and will be audio-recorded with your consent. All information you provide will remain strictly confidential and will be used exclusively for academic research.

You are welcome to decline to answer any question or withdraw from the interview at any point if you feel uncomfortable.

Instructions for the Interviewer

- Ensure that informed consent has been obtained prior to starting the interview.
- Encourage open, detailed responses by asking follow-up questions as needed.
- Reassure participants that all responses will remain confidential and used solely for research purposes.
- Record the interview (with permission) for transcription and analysis.

Interview Questions

1. Can you describe the key cybersecurity threats or risks that K2 Partnering Solutions has encountered in recent years, and how the organization responded to them?
2. What specific cybersecurity strategies, policies, or technologies has the company implemented to mitigate these threats?
3. How does K2 Partnering Solutions ensure that employees at different organizational levels are made aware of cybersecurity risks and responsibilities?
4. Can you explain how the organization designs and delivers cybersecurity training programs for staff?
5. In your view, how effective are the current training and awareness programs in changing employee behavior or reducing cyber incidents?
6. What challenges has the organization faced in implementing cybersecurity training and awareness programs?
7. How are employees' cybersecurity competencies and awareness levels assessed or evaluated after training?

8. To what extent do you think employee awareness and training contribute to the long-term sustainability and performance of the organization?
9. How does management demonstrate its commitment to cybersecurity awareness and culture across the organization?
10. What improvements or innovations would you recommend to enhance cybersecurity awareness and training at K2 Partnering Solutions?

Closing Remarks

Thank you once again for your valuable time and insights. Your participation is greatly appreciated and will contribute significantly to this academic research. Should you have any questions or require further information, please feel free to contact me.

Appendix C: Survey Questionnaire

Instructions to Participants:

Please answer the following questions based on your experience and perception of cybersecurity awareness and training within K2 Partnering Solutions.

Your responses will remain anonymous and will be used solely for academic research.

1. **How confident are you in your understanding of basic cybersecurity practices relevant to your role?**

   ☐ Very confident

   ☐ Somewhat confident

   ☐ Neutral

   ☐ Slightly unconfident

   ☐ Not confident at all

2. **To what extent has K2 Partnering Solutions provided you with adequate training on identifying and responding to cyber threats (e.g., phishing, malware, data breaches)?**

   ☐ Extensive training

   ☐ Adequate training

   ☐ Minimal training

   ☐ No training provided

   ☐ Not sure

3. **How frequently do you receive cybersecurity awareness training or updates from the organization?**

   ☐ More than once per quarter

   ☐ Once per quarter

   ☐ Twice a year

   ☐ Once a year

   ☐ Rarely or never

4. **How effective do you find the current cybersecurity training programs in helping you recognize and avoid potential cyber threats?**

   ☐ Extremely effective

   ☐ Very effective

   ☐ Moderately effective

   ☐ Slightly effective

   ☐ Not effective at all

5. **Have you ever reported a potential cybersecurity issue or threat while working at K2 Partnering Solutions?**

   ☐ Yes

   ☐ No

   ☐ I was not sure if or how to report an issue

6. **Please describe a situation (if any) where cybersecurity training or awareness helped you prevent a potential incident.**

_____

_____

_____

7. **In your opinion, how well does K2 Partnering Solutions promote a culture of cybersecurity awareness across all levels of the organization?**

☐ Very well

☐ Well

☐ Somewhat well

☐ Poorly

☐ Very poorly

8. **To what extent do you agree with the following statement: "Cybersecurity awareness and training have a direct impact on the company's long-term success and sustainability."**

☐ Strongly agree

☐ Agree

☐ Neutral

☐ Disagree

☐ Strongly disagree

9. How involved is senior management in communicating and supporting cybersecurity practices and awareness?

☐ Very involved

☐ Somewhat involved

☐ Minimally involved

☐ Not involved

☐ Not sure

10. How would you rate your current ability to identify suspicious emails or behavior that might pose a cybersecurity risk?

☐ Excellent

☐ Good

☐ Fair

☐ Poor

☐ Very poor

11. What additional cybersecurity resources or support would you find most beneficial? (Select all that apply)

☐ More frequent training sessions

☐ Simulated phishing tests

☐ Easily accessible cybersecurity guidelines or manuals

☐ Regular updates from the IT/security team

☐ One-on-one coaching or consultation

☐ Other: _____

12. What improvements would you suggest to enhance the cybersecurity awareness and training programs at K2 Partnering Solutions?

_____

_____

_____

Appendix D: Document Review Checklist

Research Focus: Cybersecurity Awareness, Training, and Business Sustainability

Organization: K2 Partnering Solutions

Researcher: Rodger Lutalo,

Doctoral Candidate,

California Intercontinental University

Date of Review: _____

A. Cybersecurity Policy and Governance

Item	Criteria	Yes / No	Notes/Comments
A.1	Presence of a documented cybersecurity policy		
A.2	Policy includes roles and responsibilities (management & employees)		
A.3	Policy outlines acceptable use, access controls, and data protection		
A.4	Policy has been updated in the last 12–18 months		
A.5	Policy is communicated to all employees and accessible		
A.6	Evidence of executive/leadership endorsement		

B. Cybersecurity Training and Awareness Programs

Item	Criteria	Yes/No	Notes/Comments
B.1	Existence of a cybersecurity training and awareness program		
B.2	Training is mandatory for all new hires (onboarding)		
B.3	Periodic (e.g., annual or quarterly) refresher training is conducted		
B.4	Training materials include phishing awareness, password hygiene, etc.		
B.5	Training is tailored for different employee roles (technical vs. non-technical)		
B.6	Training evaluation results (e.g., quizzes, feedback) are recorded		
B.7	Evidence of management participation in training sessions		

C. Cybersecurity Incident Logs and Reporting

Item	Criteria	Yes/No	Notes/Comments
C.1	Existence of an incident response policy and process		

C.2	Logs of reported incidents (e.g., phishing, malware) are maintained		
C.3	Incident logs include date, type, impact, and resolution steps		
C.4	Timeliness of response is recorded and evaluated		
C.5	Follow-up actions (e.g., retraining, system changes) are documented		
C.6	Trend analysis of incidents is conducted over time		

D. Performance and Sustainability Indicators

Item	Criteria	Yes/No	Notes/Comments
D.1	Records link training outcomes to risk reduction or performance		
D.2	Cybersecurity metrics/KPIs are used to monitor program effectiveness		
D.3	Evidence of cost savings, incident reduction, or improved compliance		
D.4	Cybersecurity practices are integrated into overall sustainability planning		

E. Compliance and Audit Documentation

Item	Criteria	Yes/No	Notes/Comments
E.1	Documentation shows compliance with relevant standards (e.g., NIST, ISO/IEC 27001)		
E.2	Internal or external audits of cybersecurity practices conducted		
E.3	Audit findings lead to specific corrective actions		
E.4	Compliance training or awareness is documented		

F. General Document Attributes

Item	Criteria	Yes/No	Notes/Comments
F.1	Documents are up to date and version-controlled		
F.2	Documents show evidence of approval and ownership		
F.3	Confidentiality and access restrictions are applied to sensitive documents		

Signature of Reviewer: _____

Date: _____